

Provenance Ledger

Concentricity

This document is generated from the author's master and fresh Lean checks. Each statement is typeset from the master. Every axiom line below is Lean's literal output.

Master revision. 1e9cc878c873cb561d9c125ec01a98a2e18333c7b1cd37798bd6d878ffde6ab8

Lean-source revision. deaa5061da811c097e4ff76abd168f516b65927d2cced98ef06ba57786250ee7

1. Theorem - Meromorphic continuation and functional equation; Riemann `\cite{Riemann1859}`

Author's statement

The Dirichlet series $\zeta(s) = \sum_{n \geq 1} n^{-s}$, convergent for $\operatorname{Re} s > 1$, continues to a meromorphic function on $\mathbb{C}$ with a single simple pole, at $s = 1$; equivalently, a holomorphic map to the Riemann sphere

$$\zeta : \mathbb{C}^* \longrightarrow \mathbb{C}^*, \quad \mathbb{C}^* = \mathbb{C} \cup \{\infty\} = S^2, \quad \zeta(1) = \infty.$$

The completed function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ is entire and satisfies

$$\xi(s) = \xi(1-s), \quad \xi(\bar{s}) = \overline{\xi(s)}.$$

Consequently, if ρ is a nontrivial zero of ζ then so are $\bar{\rho}$, $1 - \rho$, and $1 - \bar{\rho}$. The trivial zeros of ζ are at $s = -2, -4, \dots$; the nontrivial zeros lie in the strip $0 < \operatorname{Re} s < 1$. The Riemann Hypothesis asserts that every nontrivial zero has $\operatorname{Re} s = \frac{1}{2}$.

Lean declaration. `riemannZeta_meromorphicOn`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms riemannZeta_meromorphicOn
'riemannZeta_meromorphicOn' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. `riemannZeta_orderAt_one`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms riemannZeta_orderAt_one
'riemannZeta_orderAt_one' depends on axioms: [propext, Classical.choice, Quot.sound]
```

1. Theorem - Meromorphic continuation and functional equation; Riemann \cite{Riemann1859} (continued)

Lean declaration. [riemannZeta_conj](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms riemannZeta_conj
'riemannZeta_conj' depends on axioms: [proptext, Classical.choice, Quot.sound]
```

Lean declaration. [riemannZeta_one_sub_zero](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms riemannZeta_one_sub_zero
'riemannZeta_one_sub_zero' depends on axioms: [proptext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [nontrivialZero_re_mem_Ioo](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms nontrivialZero_re_mem_Ioo
'nontrivialZero_re_mem_Ioo' depends on axioms: [proptext, Classical.choice,
↪ Quot.sound]
```

Receipt import. RECEIPT_PRODUCER_REJECTED: certificate failures: ASection.northState_is_residueActionState_audit, ASection.residueActionState_north_input_audit, ASection.residueActionTransport_north_input_audit, ASection.northRelativeLoop_maps_audit, ASection.northComparison_of_parallelFaces_audit, ASection.relativeActionSquare_transport_audit, ASection.residueTotal_morphism_of_northComparison_audit

2. Theorem - Infinite Euler product; {\cite[Ch.~1]{Titchmarsh86}}

Author's statement

For $\text{Re } s > 1$,

$$\zeta(s) = \prod_{p \text{ prime}} (1 - p^{-s})^{-1} = \exp\left(\sum_p \sum_{k \geq 1} \frac{1}{k} p^{-ks}\right),$$

2. Theorem - Infinite Euler product; $\{\text{\cite[Ch.~1]{Titchmarsh86}}\}$ (continued)

the product taken over the infinitely many primes; it converges absolutely and is zero-free on $\text{Re } s > 1$.

Lean declaration. `zetaEulerLog`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms zetaEulerLog
'zetaEulerLog' depends on axioms: [proptext, Classical.choice, Quot.sound]
```

Lean declaration. `zetaC2_summable`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms zetaC2_summable
'zetaC2_summable' depends on axioms: [proptext, Classical.choice, Quot.sound]
```

Lean declaration. `zetaC2_zero_free`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms zetaC2_zero_free
'zetaC2_zero_free' depends on axioms: [proptext, Classical.choice, Quot.sound]
```

Lean declaration. `zetaC2_euler`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms zetaC2_euler
'zetaC2_euler' depends on axioms: [proptext, Classical.choice, Quot.sound]
```

Receipt import. `RECEIPT_PRODUCER.REJECTED`: certificate failures: `ASection.northState_is_residueActionState_audit`, `ASection.residueActionState_north_input_audit`, `ASection.residueActionTransport_north_input_audit`, `ASection.northRelativeLoop_maps_audit`, `ASection.northComparison_of_parallelFaces_audit`, `ASection.relativeActionSquare_transport_audit`, `ASection.residueTotal_morphism_of_northComparison_audit`

3. Theorem - Infinite Hadamard product; {\cite[Ch.~2]{Titchmarsh86}}

Author's statement

ξ is entire of order 1 with infinitely many zeros, which are exactly the nontrivial zeros $\{\rho\}$ of ζ , and admits the Hadamard factorization

$$\xi(s) = \xi(0) \prod_{\rho} \left(1 - \frac{s}{\rho}\right) e^{s/\rho},$$

the product taken over the infinitely many nontrivial zeros.

Lean declaration. [xi_entire](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms xi_entire
'xi_entire' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. [xi_zeros_eq_nontrivialZeros](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms xi_zeros_eq_nontrivialZeros
'xi_zeros_eq_nontrivialZeros' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [zeta_hadamard](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms zeta_hadamard
'zeta_hadamard' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. [riemannZeta_nontrivialZeros_infinite](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

3. Theorem - Infinite Hadamard product; {\cite[Ch.~2]{Titchmarsh86}} (continued)

```
#print axioms riemannZeta_nontrivialZeros_infinite
'riemannZeta_nontrivialZeros_infinite' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

Receipt import. **RECEIPT_PRODUCER_REJECTED:** certificate failures: ASection.northState_is_residueActionState_audit, ASection.residueActionState_north_input_audit, ASection.residueActionTransport_north_input_audit, ASection.northRelativeLoop_maps_audit, ASection.northComparison_of_parallelFaces_audit, ASection.relativeActionSquare_transport_audit, ASection.residueTotal_morphism_of_northComparison_audit

4. Corollary - Infinitude of the nontrivial zeros

Author's statement

ζ has infinitely many nontrivial zeros.

Lean declaration. `riemannZeta_nontrivialZeros_infinite`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms riemannZeta_nontrivialZeros_infinite
'riemannZeta_nontrivialZeros_infinite' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

Receipt import. **RECEIPT_PRODUCER_REJECTED:** certificate failures: ASection.northState_is_residueActionState_audit, ASection.residueActionState_north_input_audit, ASection.residueActionTransport_north_input_audit, ASection.northRelativeLoop_maps_audit, ASection.northComparison_of_parallelFaces_audit, ASection.relativeActionSquare_transport_audit, ASection.residueTotal_morphism_of_northComparison_audit

5. Theorem - Hardy {\cite{Hardy14}}

Author's statement

Infinitely many nontrivial zeros of ζ lie on the line $\operatorname{Re} s = \frac{1}{2}$.

Lean declaration. `zeta_criticalLine_zeros_infinite`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted with sorryAx; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms zeta_criticalLine_zeros_infinite
```

5. Theorem - Hardy \cite{Hardy14} (continued)

'zeta_criticalLine_zeros_infinite' depends on axioms: [propext, sorryAx,
↪ Classical.choice, Quot.sound]

Receipt import. **RECEIPT_PRODUCER_REJECTED:** certificate failures: ASection.northState_is_residueActionState_audit, ASection.residueActionState_north_input_audit, ASection.residueActionTransport_north_input_audit, ASection.northRelativeLoop_maps_audit, ASection.northComparison_of_parallelFaces_audit, ASection.relativeActionSquare_transport_audit, ASection.residueTotal_morphism_of_northComparison_audit

6. Definition - Compactified classical zeta

Author's statement

$\zeta_{\mathbb{C}^*} : \mathbb{C}^* \rightarrow \mathbb{C}^*$ is the meromorphic continuation of ζ regarded as a map to the Riemann sphere: $\zeta_{\mathbb{C}^*}(s) = \zeta(s)$ for $s \in \mathbb{C} \setminus \{1\}$; $\zeta_{\mathbb{C}^*}(1) = \infty$ (the simple pole, Theorem 1.1); and $\zeta_{\mathbb{C}^*}(\infty) = 1$, since $\zeta(s) \rightarrow 1$ as $\text{Re}(s) \rightarrow +\infty$ (the p -test: only the $n = 1$ term of $\sum_n n^{-s}$ survives). It is holomorphic into $\mathbb{C}^*$ on $\mathbb{C}^* \setminus \{1\}$, its sole pole being $s = 1$. The functional equation, the conjugate symmetry $\zeta_{\mathbb{C}^*}(\bar{s}) = \zeta_{\mathbb{C}^*}(s)$, and Hardy's infinitude (Theorem 1.5) hold verbatim for $\zeta_{\mathbb{C}^*}$, as they concern values on $\mathbb{C} \setminus \{1\}$ where $\zeta_{\mathbb{C}^*} = \zeta$.

Lean declaration. [zetaC](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms zetaC
'zetaC' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. [zetaC_coe](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms zetaC_coe
'zetaC_coe' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. [zetaC_one](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

6. Definition - Compactified classical zeta (continued)

```
#print axioms zetaC_one
'zetaC_one' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. `zetaC_infty`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms zetaC_infty
'zetaC_infty' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Receipt import. `RECEIPT_PRODUCER_REJECTED`: certificate failures: `ASection.northState_is_residueActionState_audit`, `ASection.residueActionState_north_input_audit`, `ASection.residueActionTransport_north_input_audit`, `ASection.northRelativeLoop_maps_audit`, `ASection.northComparison_of_parallelFaces_audit`, `ASection.relativeActionSquare_transport_audit`, `ASection.residueTotal_morphism_of_northComparison_audit`

7. Lemma - Compactification preserves the zero set

Author's statement

$Z(\zeta_{\mathbb{C}^*}) = Z(\zeta)$: the zeros of $\zeta_{\mathbb{C}^*}$ in $\mathbb{C}^*$ are exactly the zeros of ζ in $\mathbb{C}$; in particular the nontrivial zeros coincide.

Lean declaration. `zetaC_coe_eq_zero_iff`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms zetaC_coe_eq_zero_iff
'zetaC_coe_eq_zero_iff' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. `zetaC_zero_iff`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms zetaC_zero_iff
'zetaC_zero_iff' depends on axioms: [propext, Classical.choice, Quot.sound]
```

7. Lemma - Compactification preserves the zero set (continued)

Receipt import. `RECEIPT_PRODUCER_REJECTED`: certificate failures: `ASection.northState_is_residueActionState_audit`, `ASection.residueActionState_north_input_audit`, `ASection.residueActionTransport_north_input_audit`, `ASection.northRelativeLoop_maps_audit`, `ASection.northComparison_of_parallelFaces_audit`, `ASection.relativeActionSquare_transport_audit`, `ASection.residueTotal_morphism_of_northComparison_audit`

8. Definition

Author's statement

The octonions are: (i) a *division algebra*: every nonzero element has a multiplicative inverse; (ii) a *normed algebra*: $|ab| = |a||b|$; (iii) *non-associative*: $(ab)c \neq a(bc)$ in general; (iv) *alternative*: $(aa)b = a(ab)$ and $a(bb) = (ab)b$.

Lean declaration. `Octonion`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms Octonion
'Octonion' depends on axioms: [proptext, Classical.choice, Quot.sound]
```

Lean declaration. `Octonion.normSq_mul`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms Octonion.normSq_mul
'Octonion.normSq_mul' depends on axioms: [proptext, Classical.choice, Quot.sound]
```

Lean declaration. `Octonion.alt_left`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms Octonion.alt_left
'Octonion.alt_left' depends on axioms: [proptext, Classical.choice, Quot.sound]
```

Lean declaration. `Octonion.alt_right`

Author's object.

8. Definition (continued)

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms Octonion.alt_right
'Octonion.alt_right' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Receipt import. RECEIPT_PRODUCER_REJECTED: certificate failures: ASection.northState_is_residueActionState_audit, ASection.residueActionState_north_input_audit, ASection.residueActionTransport_north_input_audit, ASection.northRelativeLoop_maps_audit, ASection.northComparison_of_parallelFaces_audit, ASection.relativeActionSquare_transport_audit, ASection.residueTotal_morphism_of_northComparison_audit

9. Theorem - Artin \cite{Schafer66}

Author's statement

In an alternative algebra, the subalgebra generated by any two elements is associative.

No exact declaration/object check is available for this clause.

Receipt import. RECEIPT_PRODUCER_REJECTED: certificate failures: ASection.northState_is_residueActionState_audit, ASection.residueActionState_north_input_audit, ASection.residueActionTransport_north_input_audit, ASection.northRelativeLoop_maps_audit, ASection.northComparison_of_parallelFaces_audit, ASection.relativeActionSquare_transport_audit, ASection.residueTotal_morphism_of_northComparison_audit

10. Corollary

Author's statement

Powers s^n are well-defined for any $s \in \mathbb{O}$.

No exact declaration/object check is available for this clause.

Receipt import. RECEIPT_PRODUCER_REJECTED: certificate failures: ASection.northState_is_residueActionState_audit, ASection.residueActionState_north_input_audit, ASection.residueActionTransport_north_input_audit, ASection.northRelativeLoop_maps_audit, ASection.northComparison_of_parallelFaces_audit, ASection.relativeActionSquare_transport_audit, ASection.residueTotal_morphism_of_northComparison_audit

11. Definition - Complex slices and slice Riemann spheres

Author's statement

For any unit imaginary octonion $v \in S^6$, the *complex slice* is $\mathbb{C}_v = \{a + bv : a, b \in \mathbb{R}\} \subset \mathbb{O}$. Since $v^2 = -1$, this is a subalgebra isomorphic to $\mathbb{C}$ via the $\mathbb{R}$ -algebra isomorphism

11. Definition - Complex slices and slice Riemann spheres (continued)

$\varphi_v : \mathbb{C} \rightarrow \mathbb{C}_v$, $i \mapsto v$. All slices share the real axis: $\mathbb{C}_v \cap \mathbb{C}_w = \mathbb{R}$ for $v \neq \pm w$. Its one-point compactification is the *slice Riemann sphere* $\mathbb{C}_v^* := \mathbb{C}_v \cup \{\infty\} = S_v^2$, and φ_v extends to an isomorphism of Riemann spheres $\varphi_v : \mathbb{C}^* \xrightarrow{\sim} \mathbb{C}_v^*$ with $\varphi_v(\infty) = \infty$ (GPV [17], Prop. 4.1/Thm. 4.2). All slice spheres share the single point at infinity, as well as the real axis.

Lean declaration. [Octonion.unitImaginarySphere](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms Octonion.unitImaginarySphere
'Octonion.unitImaginarySphere' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [Octonion.sliceEmbed](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms Octonion.sliceEmbed
'Octonion.sliceEmbed' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. [Octonion.sliceEmbed_mul](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms Octonion.sliceEmbed_mul
'Octonion.sliceEmbed_mul' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. [Octonion.sliceSphere](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms Octonion.sliceSphere
'Octonion.sliceSphere' depends on axioms: [propext, Classical.choice, Quot.sound]
```

11. Definition - Complex slices and slice Riemann spheres (continued)

Receipt import. RECEIPT_PRODUCER_REJECTED: certificate failures: ASection.northState_is_residueActionState_audit, ASection.residueActionState_north_input_audit, ASection.residueActionTransport_north_input_audit, ASection.northRelativeLoop_maps_audit, ASection.northComparison_of_parallelFaces_audit, ASection.relativeActionSquare_transport_audit, ASection.residueTotal_morphism_of_northComparison_audit

12. Definition - Slice regularity; {\cite[Def.~5.1.1]{CSS12}}

Author's statement

Let $\Omega \subseteq \mathbb{O}$ be open. A function $f : \Omega \rightarrow \mathbb{O}$ is *slice regular* if for each $v \in S^6$, $\varphi_v^{-1} \circ f \circ \varphi_v$ satisfies the Cauchy–Riemann equations on $\varphi_v^{-1}(\Omega \cap \mathbb{C}_v) \subseteq \mathbb{C}$. A domain Ω is *axially symmetric* if $\sigma + \gamma v \in \Omega$ implies $\sigma + \gamma w \in \Omega$ for all $w \in S^6$.

No exact declaration/object check is available for this clause.

Receipt import. RECEIPT_PRODUCER_REJECTED: certificate failures: ASection.northState_is_residueActionState_audit, ASection.residueActionState_north_input_audit, ASection.residueActionTransport_north_input_audit, ASection.northRelativeLoop_maps_audit, ASection.northComparison_of_parallelFaces_audit, ASection.relativeActionSquare_transport_audit, ASection.residueTotal_morphism_of_northComparison_audit

13. Theorem - Representation Formula; {\cite[Thm.~5.1.7]{CSS12}}

Author's statement

Let f be slice regular on an axially symmetric slice domain Ω . Fix $v_0 \in S^6$ and write $f(\sigma + \gamma v_0) = \alpha(\sigma, \gamma) + v_0 \cdot \beta(\sigma, \gamma)$ with $\alpha, \beta : \mathbb{R}^2 \rightarrow \mathbb{R}$. Then for every $v \in S^6$, $f(\sigma + \gamma v) = \alpha(\sigma, \gamma) + v \cdot \beta(\sigma, \gamma)$, where $\alpha = \frac{1}{2}[f(\sigma + \gamma v_0) + f(\sigma - \gamma v_0)]$ and $\beta = \frac{1}{2}v_0^{-1}[f(\sigma - \gamma v_0) - f(\sigma + \gamma v_0)]$.

No exact declaration/object check is available for this clause.

Receipt import. RECEIPT_PRODUCER_REJECTED: certificate failures: ASection.northState_is_residueActionState_audit, ASection.residueActionState_north_input_audit, ASection.residueActionTransport_north_input_audit, ASection.northRelativeLoop_maps_audit, ASection.northComparison_of_parallelFaces_audit, ASection.relativeActionSquare_transport_audit, ASection.residueTotal_morphism_of_northComparison_audit

14. Theorem - Identity Theorem; {\cite[Cor.~5.1.9]{CSS12}}

Author's statement

Let f be slice regular on an axially symmetric slice domain Ω . If f vanishes on a subset of a single slice $\Omega \cap \mathbb{C}_{v_0}$ with an accumulation point in $\Omega \cap \mathbb{C}_{v_0}$, then $f \equiv 0$ on all of Ω .

Lean declaration. `stem_identity`

Author's object.

14. Theorem - Identity Theorem; $\{\text{\cite[Cor.~5.1.9]{CSS12}}\}$ (continued)

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unrati ed

Lean command and literal axiom print.

```
#print axioms stem_identity
'stem_identity' depends on axioms: [propxt, Classical.choice, Quot.sound]
```

Receipt import. RECEIPT_PRODUCER_REJECTED: certificate failures: ASection.northState_is_residueActionState_audit, ASection.residueActionState_north_input_audit, ASection.residueActionTransport_north_input_audit, ASection.northRelativeLoop_maps_audit, ASection.northComparison_of_parallelFaces_audit, ASection.relativeActionSquare_transport_audit, ASection.residueTotal_morphism_of_northComparison_audit

15. Theorem - Extension Theorem; $\{\text{\cite[Thm.~5.1.5]{CSS12}}\}$

Author's statement

Let $U \subseteq \mathbb{C}$ be a domain symmetric under conjugation and $g : U \rightarrow \mathbb{C}$ holomorphic. There is a unique slice regular $\text{ext}(g) : \Omega_U \rightarrow \mathbb{O}$ on $\Omega_U = \{\sigma + \gamma v : \sigma + i\gamma \in U, v \in S^6\}$ with $\text{ext}(g)|_{\mathbb{C}_{v_0}} = \varphi_{v_0} \circ g \circ \varphi_{v_0}^{-1}$ for every v_0 .

No exact declaration/object check is available for this clause.

Receipt import. RECEIPT_PRODUCER_REJECTED: certificate failures: ASection.northState_is_residueActionState_audit, ASection.residueActionState_north_input_audit, ASection.residueActionTransport_north_input_audit, ASection.northRelativeLoop_maps_audit, ASection.northComparison_of_parallelFaces_audit, ASection.relativeActionSquare_transport_audit, ASection.residueTotal_morphism_of_northComparison_audit

16. Definition - Slice-preserving functions; $\backslash\text{protect}\{\text{\cite[Def.~2.7, Rem.~2.8]{AdF}}; \text{\cite{VS, SeriesExp, Wang}}\}$

Author's statement

A slice-regular function f on an axially symmetric domain Ω is *slice preserving* if it maps every slice into itself: $f(\Omega \cap \mathbb{C}_v) \subseteq \mathbb{C}_v$ for all $v \in S^6$. For octonions ($\#S^6 > 2$) the following are equivalent characterisations:

- (i) f is *intrinsic*: $f(q^c) = f(q)^c$ for all $q \in \Omega$, where q^c is octonionic conjugation (on a slice, $a + bv \mapsto a - bv$);
- (ii) its representation-formula components are $\mathbb{R}$ -valued: writing $f(\sigma + \gamma v) = \alpha(\sigma, \gamma) + v\beta(\sigma, \gamma)$ (Theorem 3.2), one has $\alpha, \beta : \mathbb{R}^2 \rightarrow \mathbb{R}$, equivalently the inducing stem function $F = F_1 + \iota F_2$ has $\mathbb{R}$ -valued components F_1, F_2 ;
- (iii) f preserves two distinct slices; equivalently $f(\Omega \cap \mathbb{R}) \subseteq \mathbb{R}$ when Ω is a slice domain.

16. Definition - Slice-preserving functions; \protect{\cite[Def.~2.7, Rem.~2.8]{AdF}; \cite{VS, SeriesExp, Wang}} (continued)

Slice-preserving functions are the more restrictive subclass of slice-regular functions on which classical complex methods apply slicewise. They are precisely the sections of $\mathcal{R}$ (Definition 5.1), and on them the regular product is the ordinary pointwise product, so the product is commutative (Theorem 5.2; Ghiloni–Perotti–Stoppato [7, Rem. 1.8]).

Lean declaration. `IsIntrinsic`

Author’s object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms IsIntrinsic
'IsIntrinsic' depends on axioms: [propxt, Classical.choice, Quot.sound]
```

Lean declaration. `StemRing.isIntrinsic`

Author’s object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms StemRing.isIntrinsic
'StemRing.isIntrinsic' depends on axioms: [propxt, Classical.choice, Quot.sound]
```

Lean declaration. `StemRing.real_on_real`

Author’s object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms StemRing.real_on_real
'StemRing.real_on_real' depends on axioms: [propxt, Classical.choice, Quot.sound]
```

Receipt import. `RECEIPT_PRODUCER_REJECTED`: certificate failures: `ASection.northState_is_residueActionState_audit`, `ASection.residueActionState_north_input_audit`, `ASection.residueActionTransport_north_input_audit`, `ASection.northRelativeLoop_maps_audit`, `ASection.northComparison_of_parallelFaces_audit`, `ASection.relativeActionSquare_transport_audit`, `ASection.residueTotal_morphism_of_northComparison_audit`

17. Definition - Octonionic zeta

Author's statement

Here N denotes the *north pole* of S^8 : the single point at infinity ∞ adjoined in the one-point compactification $\mathbb{O}^* = \mathbb{O} \cup \{\infty\} \cong S^8$ (inverse stereographic projection; GPV [17]). It is one point with two names, so $\infty = N$; it is the only point at infinity, shared by $\mathbb{O}^*$ and by every slice Riemann sphere $\mathbb{C}_v^* = S_v^2$.

The *octonionic zeta function* $\zeta_{\mathbb{O}^*} : \mathbb{O}^* \rightarrow \mathbb{O}^*$ is defined slicewise from the compactified classical zeta (Definition 1.6) through the slice identifications $\varphi_v : \mathbb{C}^* \xrightarrow{\sim} \mathbb{C}_v^* = S_v^2$ (the $\mathbb{R}$ -algebra isomorphism $\mathbb{C} \rightarrow \mathbb{C}_v$ of Definition 2.4, extended to the Riemann spheres by $\varphi_v(\infty) = N$; GPV [17], Prop. 4.1/Thm. 4.2):

- (i) for $s \in \mathbb{O} \setminus \mathbb{R}$, with $w = \text{Im}(s)$, $v = w/|w| \in S^6$, $\gamma = |w| > 0$, $\sigma = \text{Re}(s)$, so $s = \sigma + \gamma v$:
 $\zeta_{\mathbb{O}^*}(s) := \varphi_v(\zeta_{\mathbb{C}^*}(\sigma + i\gamma))$;
- (ii) for $s \in \mathbb{R} \setminus \{1\}$: $\zeta_{\mathbb{O}^*}(s) := \zeta_{\mathbb{C}^*}(s) = \zeta(s) \in \mathbb{R}$;
- (iii) for $s = 1$ (the simple pole, Definition 1.6): $\zeta_{\mathbb{O}^*}(1) := \infty = N$;
- (iv) for $s = \infty = N$: $\zeta_{\mathbb{O}^*}(\infty) := \zeta_{\mathbb{C}^*}(\infty) = 1$.

Lean declaration. [zeta0](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms zeta0
'zeta0' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. [zeta0_infty](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms zeta0_infty
'zeta0_infty' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. [zeta0_one](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms zeta0_one
'zeta0_one' depends on axioms: [propext, Classical.choice, Quot.sound]
```

17. Definition - Octonionic zeta (continued)

Lean declaration. `zeta0_sliceEmbed`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms zeta0_sliceEmbed
'zeta0_sliceEmbed' depends on axioms: [proptext, Classical.choice, Quot.sound]
```

Lean declaration. `zeta0_sliceEmbed_of_im_neg`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms zeta0_sliceEmbed_of_im_neg
'zeta0_sliceEmbed_of_im_neg' depends on axioms: [proptext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. `zeta0_ofReal`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms zeta0_ofReal
'zeta0_ofReal' depends on axioms: [proptext, Classical.choice, Quot.sound]
```

Receipt import. `RECEIPT_PRODUCER_REJECTED`: certificate failures: `ASection.northState_is_residueActionState_audit`, `ASection.residueActionState_north_input_audit`, `ASection.residueActionTransport_north_input_audit`, `ASection.northRelativeLoop_maps_audit`, `ASection.northComparison_of_parallelFaces_audit`, `ASection.relativeActionSquare_transport_audit`, `ASection.residueTotal_morphism_of_northComparison_audit`

18. Proposition - Well-definedness

Author's statement

$\zeta_{\mathbb{O}^*} : \mathbb{O}^* \rightarrow \mathbb{O}^*$ is well-defined.

Lean declaration. `zeta0_sliceEmbed'`

Author's object.

18. Proposition - Well-definedness (continued)

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms zeta0_sliceEmbed'  
'zeta0_sliceEmbed' depends on axioms: [propeXt, Classical.choice, Quot.sound]
```

Lean declaration. [riemannZeta_conj](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms riemannZeta_conj  
'riemannZeta_conj' depends on axioms: [propeXt, Classical.choice, Quot.sound]
```

Receipt import. RECEIPT_PRODUCER_REJECTED: certificate failures: ASection.northState_is_residueActionState_audit, ASection.residueActionState_north_input_audit, ASection.residueActionTransport_north_input_audit, ASection.northRelativeLoop_maps_audit, ASection.northComparison_of_parallelFaces_audit, ASection.relativeActionSquare_transport_audit, ASection.residueTotal_morphism_of_northComparison_audit

19. Definition - The ring of slice-preserving functions under the regular product

Author's statement

$\mathcal{R} := \mathcal{O}_{\mathbb{O}^*}$ is the set of all slice-regular functions $f : \mathbb{O}^* \rightarrow \mathbb{O}^*$ that are slice preserving (Definition 3.5), $f(\Omega_v) \subseteq \mathbb{C}_v^*$ for all $v \in S^6$, equipped with pointwise addition and the regular $(*)$ product.

Lean declaration. [StemRing](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms StemRing  
'StemRing' depends on axioms: [propeXt, Classical.choice, Quot.sound]
```

Receipt import. RECEIPT_PRODUCER_REJECTED: certificate failures: ASection.northState_is_residueActionState_audit, ASection.residueActionState_north_input_audit, ASection.residueActionTransport_north_input_audit, ASection.northRelativeLoop_maps_audit, ASection.northComparison_of_parallelFaces_audit, ASection.relativeActionSquare_transport_audit, ASection.residueTotal_morphism_of_northComparison_audit

20. Theorem - Regular product on slice-preserving functions; $\{\text{\cite[Rem.~2.11]{Wang}}\}$

Author's statement

For slice-preserving f, g (with $f(\Omega_v) \subseteq \mathbb{C}_v$), the regular product restricts on each slice to the pointwise product, $(f * g)|_{\Omega_v}(w) = f(w) \cdot g(w) \in \mathbb{C}_v$, and $f * g = g * f$, $(f * g) * h = f * (g * h)$ (via Artin, Theorem 2.2).

No exact declaration/object check is available for this clause.

Receipt import. `RECEIPT_PRODUCER.REJECTED`: certificate failures: `ASec-tion.northState_is_residueActionState_audit`, `ASection.residueActionState_north_input_audit`, `ASection.residueActionTransport_north_input_audit`, `ASection.northRelativeLoop_maps_audit`, `ASection.northComparison_of_parallelFaces_audit`, `ASection.relativeActionSquare_transport_audit`, `ASection.residueTotal_morphism_of_northComparison_audit`

21. Proposition

Author's statement

$(\mathcal{R}, +, *)$ is a commutative ring with identity $1 \in \mathcal{R}$.

Lean declaration. `StemRing`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms StemRing
'StemRing' depends on axioms: [propxt, Classical.choice, Quot.sound]
```

Receipt import. `RECEIPT_PRODUCER.REJECTED`: certificate failures: `ASec-tion.northState_is_residueActionState_audit`, `ASection.residueActionState_north_input_audit`, `ASection.residueActionTransport_north_input_audit`, `ASection.northRelativeLoop_maps_audit`, `ASection.northComparison_of_parallelFaces_audit`, `ASection.relativeActionSquare_transport_audit`, `ASection.residueTotal_morphism_of_northComparison_audit`

22. Proposition - $\mathcal{R}$ is an integral domain

Author's statement

If $f, g \in \mathcal{R}$ and $f * g = 0$, then $f = 0$ or $g = 0$.

Lean declaration. `StemRing.eq_zero_or_eq_zero_of_mul_eq_zero`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

22. Proposition - $\mathcal{R}$ is an integral domain (continued)

Lean command and literal axiom print.

```
#print axioms StemRing.eq_zero_or_eq_zero_of_mul_eq_zero
'StemRing.eq_zero_or_eq_zero_of_mul_eq_zero' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

Receipt import. `RECEIPT_PRODUCER.REJECTED`: certificate failures: `ASection.northState_is_residueActionState_audit`, `ASection.residueActionState_north_input_audit`, `ASection.residueActionTransport_north_input_audit`, `ASection.northRelativeLoop_maps_audit`, `ASection.northComparison_of_parallelFaces_audit`, `ASection.relativeActionSquare_transport_audit`, `ASection.residueTotal_morphism_of_northComparison_audit`

23. Theorem - The slice exponential; $\text{\protect{\cite[\S3]{GPVwind}}}$; $\text{\cite[Prop.~5.1]{VS}};$ $\text{\cite[Rem.~2.23]{AdF}}$

Author's statement

The exponential $\exp : \mathbb{O} \rightarrow \mathbb{O} \setminus \{0\}$, $\exp(q) = \sum_{k \geq 0} q^k / k!$ (powers well defined by Corollary 2.3), is a slice-regular, slice-preserving entire function (Definition 3.5), given on each slice by

$$\exp(x + Iy) = e^x (\cos y + I \sin y), \quad x, y \in \mathbb{R}, \quad I \in S^6.$$

Consequently $|\exp q| = e^{\operatorname{Re} q}$ depends only on the real part, and $\exp$ is nowhere zero.

Lean declaration. `Octonion.exp`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms Octonion.exp
'Octonion.exp' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. `Octonion.exp_sliceEmbed`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms Octonion.exp_sliceEmbed
'Octonion.exp_sliceEmbed' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. `Octonion.norm_exp`

Author's object.

Object relation. Object relation not checked

23. Theorem - The slice exponential; `\protect{\cite[\S3]{GPVwind}; \cite[Prop.~5.1]{VS}; \cite[Rem.~2.23]{AdF}}` (continued)

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms Octonion.norm_exp
'Octonion.norm_exp' depends on axioms: [proptext, Classical.choice, Quot.sound]
```

Lean declaration. `Octonion.exp_ne_zero`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms Octonion.exp_ne_zero
'Octonion.exp_ne_zero' depends on axioms: [proptext, Classical.choice, Quot.sound]
```

Receipt import. `RECEIPT_PRODUCER_REJECTED:` certificate failures: `ASection.northState_is_residueActionState_audit`, `ASection.residueActionState_north_input_audit`, `ASection.residueActionTransport_north_input_audit`, `ASection.northRelativeLoop_maps_audit`, `ASection.northComparison_of_parallelFaces_audit`, `ASection.relativeActionSquare_transport_audit`, `ASection.residueTotal_morphism_of_northComparison_audit`

24. Theorem - The logarithm manifold `$E^{+}_{\mathbb{O}}$;` `\protect{\cite[Prop.~5.1, Rem.~5.2, Def.~5.3, Prop.~5.4, Def.~5.5]{VS}; \cite[\S3]{GPVwind}}`

Author's statement

Let $T : \mathbb{O} \rightarrow \mathbb{O} \times \text{Im } \mathbb{O}$, $T(x + Iy) = (\sinh x \cos y + I \sinh x \sin y, Iy)$, and $\mathbb{O}^+ = \{q : \text{Re } q > 0\}$. The *logarithm manifold* is $E_{\mathbb{O}}^+ := T(\mathbb{O}^+)$, the semi-helicoidal hypercomplex 8-manifold; equivalently it is the image of the exponential graph, parametrised by the $E_{\mathbb{O}}^+$ -*exponential*

$$E : \mathbb{O} \longrightarrow E_{\mathbb{O}}^+ \subset \mathbb{O} \times \text{Im } \mathbb{O}, \quad E(x + Iy) = (\exp(x + Iy), Iy) = (e^x \cos y + Ie^x \sin y, Iy),$$

an immersion and a diffeomorphism, with inverse the $E_{\mathbb{O}}^+$ -*logarithm* $L : E_{\mathbb{O}}^+ \rightarrow \mathbb{O}$, $L(q, p) = \log |q| + p$ (here $p = \text{Arg}(q)$). Writing π for the projection to the first factor, $\pi \circ E = \exp$ (Theorem 6.1). The manifold $E_{\mathbb{O}}^+$ is an adapted blow-up of $\mathbb{O}$ along the real axis. Its projection $\pi : E_{\mathbb{O}}^+ \rightarrow \mathbb{O} \setminus \{0\}$ has the degenerate real fibres responsible for the failure of covering and openness. The blow-up unwraps the multivalued logarithm into the single-valued L , and a branch of the hypercomplex logarithm is $\log_{\mathbb{O}} = L \circ (\pi|_{\Omega})^{-1}$ on any path-connected $\Omega \subset E_{\mathbb{O}}^+$ on which $\pi|_{\Omega}$ is injective.

24. Theorem - The logarithm manifold $E^{\{+\}_{\infty}}$;
 $\backslash\text{protect}\{\backslash\text{cite}[\text{Prop.~5.1, Rem.~5.2, Def.~5.3, Prop.~5.4, Def.~5.5}]\{\text{VS};$
 $\backslash\text{cite}[\text{\S3}]\{\text{GPVwind}\}\}$ (continued)

Lean declaration. [Octonion.Eexp](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms Octonion.Eexp
'Octonion.Eexp' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. [Octonion.logManifold](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms Octonion.logManifold
'Octonion.logManifold' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. [Octonion.Llog](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms Octonion.Llog
'Octonion.Llog' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. [Octonion.Llog_Eexp](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms Octonion.Llog_Eexp
'Octonion.Llog_Eexp' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. [Octonion.Eexp_Llog](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

24. Theorem - The logarithm manifold $E^{\{+\}_{\infty}}$;
 $\backslash\text{protect}\{\backslash\text{cite}[\text{Prop.~5.1, Rem.~5.2, Def.~5.3, Prop.~5.4, Def.~5.5}]\{\text{VS};$
 $\backslash\text{cite}[\text{S3}]\{\text{GPVwind}\}\}$ (continued)

```
#print axioms Octonion.Eexp_Llog
'Octonion.Eexp_Llog' depends on axioms: [propept, Classical.choice, Quot.sound]
```

Lean declaration. `Octonion.exp_Llog`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms Octonion.exp_Llog
'Octonion.exp_Llog' depends on axioms: [propept, Classical.choice, Quot.sound]
```

Receipt import. `RECEIPT_PRODUCER_REJECTED: certificate failures: ASection.northState_is_residueActionState_audit, ASection.residueActionState_north_input_audit, ASection.residueActionTransport_north_input_audit, ASection.northRelativeLoop_maps_audit, ASection.northComparison_of_parallelFaces_audit, ASection.relativeActionSquare_transport_audit, ASection.residueTotal_morphism_of_northComparison_audit`

25. Lemma - The degenerate set of the exponential; its fibre over a real value

Author's statement

Sourced. With π the first-factor projection, $\pi \circ E = \exp$ ([17, Rem. 5.2(a)], the commuting triangle). Its spherical *degenerate set* makes the projection an adapted blow-up with failures of covering and openness ([17, Rem. 5.2(b)]). The direction $I(q)$ becomes singular at every point of $\mathbb{R}$ ([21, Rem. 2.1]).

Proved from the slice form. For $r > 0$,

$$\exp^{-1}(-r) = \{ \log r + I(2k+1)\pi : I \in S^6, k \in \mathbb{Z} \} :$$

by Theorem 6.1, $\exp(x + Iy) = e^x \cos y + Ie^x \sin y = -r$ forces $e^x \sin y = 0$, hence $y \in \pi\mathbb{Z}$; negativity of the value forces $y = (2k+1)\pi$ and $x = \log r$; the direction I is unconstrained. The fibre is thus indexed by the single real *level* $\log r = \log |-r|$ and, within it, by the odd winding indices $2k+1$ carried as band data (the winding lift, Theorem 6.5; [21, Cor. 5.21]). The fibre formula itself appears, stated without proof as Preface motivation, in [17] (p. 972); the slice-form derivation above is the load-bearing statement.

Lean declaration. `Octonion.exp_fibre_neg_real`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

25. Lemma - The degenerate set of the exponential; its fibre over a real value (continued)

```
#print axioms Octonion.exp_fibre_neg_real
'Octonion.exp_fibre_neg_real' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. `Octonion.exp_eq_neg_ofReal_iff`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms Octonion.exp_eq_neg_ofReal_iff
'Octonion.exp_eq_neg_ofReal_iff' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. `Octonion.logManifold_fibre_neg_real`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms Octonion.logManifold_fibre_neg_real
'Octonion.logManifold_fibre_neg_real' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. `Octonion.degenerate_level_readout`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms Octonion.degenerate_level_readout
'Octonion.degenerate_level_readout' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Receipt import. `RECEIPT_PRODUCER.REJECTED`: certificate failures: `ASection.northState_is_residueActionState_audit`, `ASection.residueActionState_north_input_audit`, `ASection.residueActionTransport_north_input_audit`, `ASection.northRelativeLoop_maps_audit`, `ASection.northComparison_of_parallelFaces_audit`, `ASection.relativeActionSquare_transport_audit`, `ASection.residueTotal_morphism_of_northComparison_audit`

26. Definition - Slice restriction and the I -independent lift;
 $\text{\cite[Rem.~2.11]{Wang}}; \quad \text{\cite[\S3.3,~\S3.7]{BisiWinkelmann}};$
 $\text{\cite[Def.~2.1, Prop.~2.2]{SeriesExp}}$

Author's statement

A slice-preserving section A carries each slice Riemann sphere into itself,

$$A(\mathbb{C}_v^*) \subseteq \mathbb{C}_v^* = S_v^2 \quad (v \in S^6),$$

so its restriction to a slice is a holomorphic self-map of that slice sphere,

$$A_v := \varphi_v^{-1} \circ A \circ \varphi_v : \mathbb{C}^* \longrightarrow \mathbb{C}^*,$$

through the identification $\varphi_v : \mathbb{C}^* \xrightarrow{\sim} \mathbb{C}_v^*$ of Definition 2.4. The *I-independent lift* is the fact that this holomorphic stem is the *same* for every v : there is one holomorphic $A_o : \mathbb{C}^* \rightarrow \mathbb{C}^*$ with

$$A|_{\mathbb{C}_v^*} = \varphi_v \circ A_o \circ \varphi_v^{-1} \quad \text{for all } v \in S^6$$

(Wang [16, Rem. 2.11]; Bisi–Winkelmann [4, §3.7]). Equivalently, A is *intrinsic*, $A(q^c) = A(q)^c$, and is induced ($A = \mathcal{I}(F)$) by a single stem function $F = F_1 + \iota F_2 : D \rightarrow \mathbb{O}_{\mathbb{C}}$ on the symmetric $D \subset \mathbb{C}$ whose components F_1, F_2 are $\mathbb{R}$ -valued (the stem-function formalism, Ghiloni–Perotti [7, Def. 2.1, Prop. 2.2]):

$$A(x + Iy) = F_1(x + iy) + I F_2(x + iy), \quad F_1, F_2 \text{ } \mathbb{R}\text{-valued.}$$

It is this single I -independent slice stem that makes a slice-preserving section G_2 -equivariant, hence blind to the G_2 -orbit direction (Definition 9.2).

Lean declaration. `IsIntrinsic`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms IsIntrinsic
'IsIntrinsic' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. `ASection.realize`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.realize
'ASection.realize' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. `ASection.realize_mem_sliceSphere`

26. Definition - Slice restriction and the $\mathbb{I}\mathbb{S}$ -independent lift; $\backslash\text{protect}\{\backslash\text{cite}[\text{Rem.}\sim 2.11]\{\text{Wang}\}; \backslash\text{cite}[\text{S3.3},\sim\text{S3.7}]\{\text{BisiWinkelmann}\}; \backslash\text{cite}[\text{Def.}\sim 2.1, \text{Prop.}\sim 2.2]\{\text{SeriesExp}\}\}$ (continued)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.realize_mem_sliceSphere
'ASection.realize_mem_sliceSphere' depends on axioms: [proptext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [ASection.realize_equivariant](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.realize_equivariant
'ASection.realize_equivariant' depends on axioms: [proptext, Classical.choice,
↪ Quot.sound]
```

Receipt import. RECEIPT_PRODUCER_REJECTED: certificate failures: ASection.northState_is_residueActionState_audit, ASection.residueActionState_north_input_audit, ASection.residueActionTransport_north_input_audit, ASection.northRelativeLoop_maps_audit, ASection.northComparison_of_parallelFaces_audit, ASection.relativeActionSquare_transport_audit, ASection.residueTotal_morphism_of_northComparison_audit

27. Theorem - The winding lift; $\backslash\text{protect}\{\backslash\text{cite}[\text{Def.}\sim 3.4, \text{Def.}\sim 4.1, \text{Prop.}\sim 4.2, \text{Def.}\sim 5.11]\{\text{GPVwind}\}\}$

Author's statement

Let $\gamma : [a, b] \rightarrow \mathbb{O} \setminus \{0\}$ be a path (on each slice $\mathbb{C}_v$ this is the classical complex continuation). A *continuation of the hypercomplex logarithm along γ* is a path $\tilde{\gamma} : [a, b] \rightarrow \mathbb{O}$ with $\exp \circ \tilde{\gamma} = \gamma$, and a *lift of γ to the logarithm manifold $E_{\mathbb{O}}^+$* (Theorem 6.2) is a path $\Gamma : [a, b] \rightarrow E_{\mathbb{O}}^+$ with $\text{pr}_1 \circ \Gamma = \gamma$:

$$\begin{array}{ccc}
 & \nearrow \tilde{\gamma} & \mathbb{O} \\
 [a, b] & \xrightarrow{\gamma} & \mathbb{O} \setminus \{0\} \xleftarrow{\exp}
 \end{array}
 \qquad
 \begin{array}{ccc}
 & \nearrow \Gamma & E_{\mathbb{O}}^+ \\
 [a, b] & \xrightarrow{\gamma} & \mathbb{O} \setminus \{0\} \xleftarrow{\text{pr}_1}
 \end{array}$$

The two data are *equivalent*: a lift Γ exists if and only if a continuation $\tilde{\gamma}$ exists, and they correspond by

$$\tilde{\gamma} = L \circ \Gamma, \quad \Gamma = (\exp \circ \tilde{\gamma}, \text{Im } \tilde{\gamma}),$$

27. Theorem - The winding lift; \protect{\cite[Def.~3.4, Def.~4.1, Prop.~4.2, Def.~5.11]{GPVwind}} (continued)

where L is the $E_{\mathbb{O}}^+$ -logarithm of Theorem 6.2 ([21, Prop. 4.2]). For a *loop* $\gamma : S^1 \rightarrow \mathbb{O} \setminus \{0\}$ the lift is taken on the universal cover $\exp : i\mathbb{R} \rightarrow S^1$, as a continuous $\Gamma : i\mathbb{R} \rightarrow E_{\mathbb{O}}^+$ with $\text{pr}_1 \circ \Gamma = \gamma \circ \exp$ ([21, Def. 5.11]):

$$\begin{array}{ccc} i\mathbb{R} & \xrightarrow{\Gamma} & E_{\mathbb{O}}^+ \\ \exp \downarrow & & \downarrow \text{pr}_1 \\ S^1 & \xrightarrow{\gamma} & \mathbb{O} \setminus \{0\} \end{array}$$

Lean declaration. [exists_log_continuation](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms exists_log_continuation
'exists_log_continuation' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. [winding_lift_unique](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms winding_lift_unique
'winding_lift_unique' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. [Octonion.lift_iff_continuation](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms Octonion.lift_iff_continuation
'Octonion.lift_iff_continuation' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Receipt import. RECEIPT_PRODUCER_REJECTED: certificate failures: ASection.northState_is_residueActionState_audit, ASection.residueActionState_north_input_audit, ASection.residueActionTransport_north_input_audit, ASection.northRelativeLoop_maps_audit, ASection.northComparison_of_parallelFaces_audit, ASection.relativeActionSquare_transport_audit, ASection.residueTotal_morphism_of_northComparison_audit

28. Proposition - Tameness, existence, and the winding number; $\backslash\text{protect}\{\backslash\text{cite}[\text{Def.~4.20, Cor.~5.21, Cor.~5.22}]\{\text{GPVwind}\}\}$

Author's statement

The lift of Theorem 6.5 is governed by the companion structure of γ .

- (i) *Uniqueness (tameness).* γ is *tame* when it admits a *unique* companion imaginary-unit field; its lift is then unique ([21, Def. 4.20]).
- (ii) *Existence and closure.* A lift of a loop γ exists if and only if the signature satisfies $\sigma(\gamma|_{[\xi_l, \xi_{l+1}]}) \in \{0, -1\}$ on each obstruction interval; and *if it exists, the lift is itself a loop* ([21, Cor. 5.13]; pinned against the arXiv text; the earlier citation to Cor. 5.22 is superseded).
- (iii) *Winding number.* For an untwisted tame loop with even circular signature σ^c , the winding number is $\omega = |\sigma^c|/2$ ([21, Cor. 5.21]).

Lean declaration. [winding_lift_unique](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms winding_lift_unique
'winding_lift_unique' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. [winding_loop_defect](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms winding_loop_defect
'winding_loop_defect' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Receipt import. RECEIPT_PRODUCER_REJECTED: certificate failures: ASection.northState_is_residueActionState_audit, ASection.residueActionState_north_input_audit, ASection.residueActionTransport_north_input_audit, ASection.northRelativeLoop_maps_audit, ASection.northComparison_of_parallelFaces_audit, ASection.relativeActionSquare_transport_audit, ASection.residueTotal_morphism_of_northComparison_audit

29. Theorem - Slice preservation

Author's statement

$\zeta_{\mathbb{O}^*}$ is slice preserving (Definition 3.5): $\zeta_{\mathbb{O}^*}(\mathbb{C}_v^*) \subseteq \mathbb{C}_v^*$ for every $v \in S^6$.

29. Theorem - Slice preservation (continued)

Lean declaration. `zeta0_mem_sliceSphere`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms zeta0_mem_sliceSphere
'zeta0_mem_sliceSphere' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Receipt import. `RECEIPT_PRODUCER_REJECTED`: certificate failures: `ASection.northState_is_residueActionState_audit`, `ASection.residueActionState_north_input_audit`, `ASection.residueActionTransport_north_input_audit`, `ASection.northRelativeLoop_maps_audit`, `ASection.northComparison_of_parallelFaces_audit`, `ASection.relativeActionSquare_transport_audit`, `ASection.residueTotal_morphism_of_northComparison_audit`

30. Theorem - $\zeta_{\mathbb{O}^*}$ is a section of $\mathcal{R}$

Author's statement

$\zeta_{\mathbb{O}^*} \in \mathcal{R}$.

Lean declaration. `zetaSection`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms zetaSection
'zetaSection' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Receipt import. `RECEIPT_PRODUCER_REJECTED`: certificate failures: `ASection.northState_is_residueActionState_audit`, `ASection.residueActionState_north_input_audit`, `ASection.residueActionTransport_north_input_audit`, `ASection.northRelativeLoop_maps_audit`, `ASection.northComparison_of_parallelFaces_audit`, `ASection.relativeActionSquare_transport_audit`, `ASection.residueTotal_morphism_of_northComparison_audit`

31. Definition - The automorphism group G_2

Author's statement

$G_2 = \text{Aut}(\mathbb{O})$ is the 14-dimensional compact, connected, simply connected Lie group of $\mathbb{R}$ -algebra automorphisms of $\mathbb{O}$. Each $g \in G_2$ fixes 1, hence the real axis pointwise, and acts on $\text{Im } \mathbb{O} \cong \mathbb{R}^7$ as an element of $SO(7)$; thus $g \cdot (\sigma + \gamma v) = \sigma + \gamma g(v)$.

31. Definition - The automorphism group G_2 (continued)

Lean declaration. [G2](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms G2
'G2' depends on axioms: [propeXt, Classical.choice, Quot.sound]
```

Lean declaration. [G2.smul_one](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms G2.smul_one
'G2.smul_one' depends on axioms: [propeXt, Classical.choice, Quot.sound]
```

Lean declaration. [G2.smul_ofReal](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms G2.smul_ofReal
'G2.smul_ofReal' depends on axioms: [propeXt, Classical.choice, Quot.sound]
```

Receipt import. [RECEIPT_PRODUCER_REJECTED](#): certificate failures: [ASec-tion.northState.is_residueActionState_audit](#), [ASec-tion.residueActionState.north_input_audit](#), [ASec-tion.residueActionTransport.north_input_audit](#), [ASec-tion.northRelativeLoop_maps_audit](#), [ASec-tion.northComparison_of_parallelFaces_audit](#), [ASec-tion.relativeActionSquare_transport_audit](#), [ASec-tion.residueTotal_morphism_of_northComparison_audit](#)

32. Theorem - G_2 -equivariance

Author's statement

$\zeta_{\mathbb{O}^*}(g \cdot s) = g \cdot \zeta_{\mathbb{O}^*}(s)$ for all $g \in G_2$, $s \in \mathbb{O}$.

Lean declaration. [zeta0_equivariant](#)

Author's object.

Object relation. Object relation not checked

32. Theorem - $\mathbb{G}^2$ -equivariance (continued)

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms zeta0_equivariant
'zeta0_equivariant' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Receipt import. RECEIPT_PRODUCER_REJECTED: certificate failures: ASection.northState_is_residueActionState_audit, ASection.residueActionState_north_input_audit, ASection.residueActionTransport_north_input_audit, ASection.northRelativeLoop_maps_audit, ASection.northComparison_of_parallelFaces_audit, ASection.relativeActionSquare_transport_audit, ASection.residueTotal_morphism_of_northComparison_audit

33. Theorem - $\text{protect}\{\text{cite}\{\text{Baez02}\}\}$

Author's statement

G_2 acts transitively on S^6 with stabilizer $SU(3)$: $SU(3) \hookrightarrow G_2 \twoheadrightarrow S^6$.

Lean declaration. G2.exists_smul_eq_of_mem_unitImaginarySphere

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms G2.exists_smul_eq_of_mem_unitImaginarySphere
'G2.exists_smul_eq_of_mem_unitImaginarySphere' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

Receipt import. RECEIPT_PRODUCER_REJECTED: certificate failures: ASection.northState_is_residueActionState_audit, ASection.residueActionState_north_input_audit, ASection.residueActionTransport_north_input_audit, ASection.northRelativeLoop_maps_audit, ASection.northComparison_of_parallelFaces_audit, ASection.relativeActionSquare_transport_audit, ASection.residueTotal_morphism_of_northComparison_audit

34. Theorem - Zero Equivalence Theorem

Author's statement

Let $\rho = \sigma + i\gamma \in \mathbb{C}^*$. For every $v \in S^6$,

$$\zeta_{\mathbb{O}^*}(\sigma + \gamma v) = 0 \iff \zeta_{\mathbb{C}^*}(\rho) = 0.$$

Equivalently: a non-real $s \in \mathbb{O}^*$ is a zero of $\zeta_{\mathbb{O}^*}$ if and only if its slice coordinate $\varphi_v^{-1}(s)$ is a zero of $\zeta_{\mathbb{C}^*}$; by Lemma 1.7 these are exactly the nontrivial zeros of the classical ζ .

34. Theorem - Zero Equivalence Theorem (continued)

Lean declaration. `sliceEmbed_eq_zero_iff`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms sliceEmbed_eq_zero_iff
'sliceEmbed_eq_zero_iff' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. `zeta0_zero_iff`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms zeta0_zero_iff
'zeta0_zero_iff' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Receipt import. `RECEIPT_PRODUCER_REJECTED: certificate failures: ASection.northState_is_residueActionState_audit, ASection.residueActionState_north_input_audit, ASection.residueActionTransport_north_input_audit, ASection.northRelativeLoop_maps_audit, ASection.northComparison_of_parallelFaces_audit, ASection.relativeActionSquare_transport_audit, ASection.residueTotal_morphism_of_northComparison_audit`

35. Theorem - Nontrivial zeros are infinitely many disjoint S^6 -spheres

Author's statement

For a nontrivial zero $\rho = \sigma + i\gamma$ of $\zeta_{\mathbb{C}^*}$ (so $\gamma > 0$) set

$$S_\rho = \{\sigma + \gamma v : v \in S^6\} = \sigma + \gamma S^6 \subset \mathbb{O}.$$

Then:

- (i) S_ρ is a 6-sphere, the round sphere of radius γ centred at the real point σ , equivalently the G_2 -orbit of any one of its points $\sigma + \gamma v_0$ (Theorem 7.4; G_2 acts transitively on S^6 with stabiliser $SU(3)$, Theorem 7.5, Baez [3]).
- (ii) every point of S_ρ is a zero of $\zeta_{\mathbb{O}^*}$, and conjugate zeros give the same sphere, $S_\rho = S_{\bar{\rho}}$;
- (iii) the nontrivial-zero set is the disjoint union $Z_{\mathbb{O}^*}^{\text{nt}} = \bigsqcup_{[\rho]} S_\rho$ over conjugate pairs $[\rho] = \{\rho, \bar{\rho}\}$, with distinct pairs giving disjoint spheres;
- (iv) there are infinitely many such spheres (Corollary 1.4; also Hardy [11], Theorem 1.5).

35. Theorem - Nontrivial zeros are infinitely many disjoint $\mathbb{Q}$ -spheres (continued)

Lean declaration. [zeroSphere](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms zeroSphere
'zeroSphere' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. [zeroSphere_eq_orbit](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms zeroSphere_eq_orbit
'zeroSphere_eq_orbit' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. [zeroSphere_zeros](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms zeroSphere_zeros
'zeroSphere_zeros' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. [zeroSphere_disjoint](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms zeroSphere_disjoint
'zeroSphere_disjoint' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. [zeroSpheres_infinite](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

35. Theorem - Nontrivial zeros are infinitely many disjoint S^6 -spheres (continued)

```
#print axioms zeroSpheres_infinite
'zeroSpheres_infinite' depends on axioms: [proptext, Classical.choice, Quot.sound]
```

Receipt import. RECEIPT_PRODUCER_REJECTED: certificate failures: ASection.northState_is_residueActionState_audit, ASection.residueActionState_north_input_audit, ASection.residueActionTransport_north_input_audit, ASection.northRelativeLoop_maps_audit, ASection.northComparison_of_parallelFaces_audit, ASection.relativeActionSquare_transport_audit, ASection.residueTotal_morphism_of_northComparison_audit

36. Theorem - The equivalence: concentricity $\Leftrightarrow$ RH

Author's statement

The following are equivalent:

- (a) the Riemann Hypothesis;
- (b) the nontrivial-zero 6-spheres $\{S_\rho\}$ of $\zeta_{\mathbb{Q}^*}$ (Theorem 8.2) are *concentric*: they share a single common centre, necessarily a point of the real axis.

When they hold, the common centre is $\frac{1}{2}$.

Lean declaration. `riemannHypothesis_iff_concentric`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unrati ed

Lean command and literal axiom print.

```
#print axioms riemannHypothesis_iff_concentric
'riemannHypothesis_iff_concentric' depends on axioms: [proptext, Classical.choice,
↪ Quot.sound]
```

Receipt import. RECEIPT_PRODUCER_REJECTED: certificate failures: ASection.northState_is_residueActionState_audit, ASection.residueActionState_north_input_audit, ASection.residueActionTransport_north_input_audit, ASection.northRelativeLoop_maps_audit, ASection.northComparison_of_parallelFaces_audit, ASection.relativeActionSquare_transport_audit, ASection.residueTotal_morphism_of_northComparison_audit

37. Definition - The compactified octonions $\mathbb{O}\mathbb{O}^{\wedge\{*\}} = S^8$

Author's statement

Let $\mathbb{O}^* = \mathbb{O} \cup \{\infty\} = S^8$ be the Alexandroff one-point compactification of $\mathbb{O} = \mathbb{R}^8$. Concretely (GPV [17], Prop. 4.1 and Thm. 4.2, for $m = \dim_{\mathbb{R}} K \in \{4, 8\}$): the inverse stereographic projections from the north and south poles,

37. Definition - The compactified octonions $\mathbb{O}\mathbb{O}^* = S^8$ (continued)

$$f(x + Iy) = \left(\frac{2(x+Iy)}{1+x^2+y^2}, \frac{-1+x^2+y^2}{1+x^2+y^2} \right), \quad h(x + Iy) = \left(\frac{2(x-Iy)}{1+x^2+y^2}, \frac{1-x^2-y^2}{1+x^2+y^2} \right),$$

give a conformal atlas $\{(K, f), (K, h)\}$ endowing S^m with the structure of a slice (quaternionic/octonionic) Riemann manifold, with transition map $h^{-1} \circ f(q) = \bar{q}/q^2 = 1/q$, itself slice regular and slice preserving. The adjoined point is the north pole N ; this is the point at infinity. In Lean the underlying compactification is `Mathlib.Topology.Compactification.OnePoint`, $\mathbb{O}^* = \text{OnePoint } \mathbb{O}$ with $\infty = \text{OnePoint.infty}$, and the homeomorphism $\mathbb{O}^* \cong S^8$ is `onePointEquivSphereOfFinrankEq` applied to $\dim_{\mathbb{R}} \mathbb{O} = 8$ (`Mathlib.Topology.Compactification.OnePoint.Sphere`). The compactification is what makes the value of a section at its real pole a well-defined point of $\mathbb{O}^*$, the point at infinity $\infty = N$. The two views $\mathcal{H}_1$ and $\mathcal{S}_2$ are built on these compactified values (Definition 9.3); the distinct projective base $\mathcal{B}$ and direction-indexed sphere world used in the construction below are introduced in Definition 9.4.

Lean declaration. `Octonion`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms Octonion
'Octonion' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. `onePointEquivSphereOfFinrankEq`

Author's object.

Object relation. Object relation not checked

Exact state. External library dependency; approved kernel-reported axiom surface; no authored-object anchor required

Lean command and literal axiom print.

```
#print axioms onePointEquivSphereOfFinrankEq
'onePointEquivSphereOfFinrankEq' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Receipt import. `RECEIPT_PRODUCER.REJECTED`: certificate failures: `ASection.northState_is_residueActionState_audit`, `ASection.residueActionState_north_input_audit`, `ASection.residueActionTransport_north_input_audit`, `ASection.northRelativeLoop_maps_audit`, `ASection.northComparison_of_parallelFaces_audit`, `ASection.relativeActionSquare_transport_audit`, `ASection.residueTotal_morphism_of_northComparison_audit`

38. Definition - The section's slice data and equivariance

Author's statement

Let $A \in \mathcal{R}$. By the I -independent lift (Definition 6.4; Wang [16, Rem. 2.11], Bisi–Winkelmann [4, §3.7]) there is one holomorphic stem with $\mathbb{R}$ -valued components,

$$A(x + Iy) = F_1(x + iy) + I F_2(x + iy), \quad F_1, F_2 \text{ } \mathbb{R}\text{-valued,}$$

the same for every $I \in S^6$. Three consequences used below:

- (i) *Slice preservation of values.* $A(\mathbb{C}_I^*) \subseteq \mathbb{C}_I^*$ for every I : the value at a point lies on that point's own slice sphere (Definition 5.1).
- (ii) *G_2 -equivariance.* For $g \in G_2$ and non-real $q = x + Iy$,

$$A(g \cdot q) = F_1 + g(I) F_2 = g \cdot (F_1 + I F_2) = g \cdot A(q),$$

since $g \cdot (x + Iy) = x + g(I)y$ (Definition 7.3) and F_1, F_2 are real; for $q \in \mathbb{R} \cup \{N\}$ both sides are G_2 -fixed, because $A(\mathbb{R}) \subseteq \mathbb{R}$ and $A(N) \in \bigcap_I \mathbb{C}_I^* = \mathbb{R} \cup \{N\}$ by (i). (For $\zeta_{\mathbb{O}^*}$ this is Theorem 7.4; the computation above is the same one, run for an arbitrary intrinsic section.)

- (iii) *Blindness to the sphere direction.* On a G_2 -orbit $S_{(\sigma, \gamma)}$ the stem coordinates $(F_1, F_2)(\sigma + i\gamma)$ are constant; in particular if A vanishes at one point of the orbit it vanishes on the whole orbit (Lemma 10.2).

A slice-preserving section carries one holomorphic stem per slice, and the direction S^6 is traversed only by the G_2 -morphisms already present in both worlds.

Lean declaration. `ASection.realize`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.realize
'ASection.realize' depends on axioms: [proptext, Classical.choice, Quot.sound]
```

Lean declaration. `ASection.realize_mem_sliceSphere`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.realize_mem_sliceSphere
'ASection.realize_mem_sliceSphere' depends on axioms: [proptext, Classical.choice,
↪ Quot.sound]
```

38. Definition - The section's slice data and equivariance (continued)

Lean declaration. `ASection.realize_equivariant`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unrati ed

Lean command and literal axiom print.

```
#print axioms ASection.realize_equivariant
'ASection.realize_equivariant' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. `ASection.sliceCoord_smul_invariant`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unrati ed

Lean command and literal axiom print.

```
#print axioms ASection.sliceCoord_smul_invariant
'ASection.sliceCoord_smul_invariant' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Receipt import. `RECEIPT_PRODUCER_REJECTED`: certificate failures: `ASection.northState_is_residueActionState_audit`, `ASection.residueActionState_north_input_audit`, `ASection.residueActionTransport_north_input_audit`, `ASection.northRelativeLoop_maps_audit`, `ASection.northComparison_of_parallelFaces_audit`, `ASection.relativeActionSquare_transport_audit`, `ASection.residueTotal_morphism_of_northComparison_audit`

39. Definition - The octonionic and slice-value worlds

Author's statement

Recall the *translation groupoid* of an action $G \curvearrowright X$: its objects are the points of X and its morphisms $x \rightarrow y$ are the elements $g \in G$ satisfying $g \cdot x = y$. Thus $\pi_0(G \ltimes X) = X/G$ (Quillen [22, §1]; in Lean, `CategoryTheory.ActionCategory`).

The octonionic world is

$$\mathcal{H}_1 = G_2 \ltimes \mathbb{O}^*.$$

Its components are the G_2 -orbits: the compactified real fixed locus $\mathbb{R} \cup \{N\}$ and, for each $\sigma \in \mathbb{R}$ and $\gamma > 0$, the residue- $\mathbb{C}$ sphere $S_{(\sigma, \gamma)} = \{\sigma + \gamma v : v \in S^6\}$, labelled by its real centre σ and radius γ (Baez [3]; Theorem 7.5).

The slice-value world $\mathcal{S}_2$ organizes the same compactified values by their slice decomposition. Every slice contains the real axis, and

$$\mathbb{O}^* = \bigcup_{I \in S^6} \mathbb{C}_I^*, \quad \mathbb{C}_I^* \cap \mathbb{C}_J^* = \mathbb{R} \cup \{N\} \quad (J \neq \pm I).$$

Its objects are the points of $\mathbb{O}^*$. Its generating paths are the direction arrows induced by G_2

39. Definition - The octonionic and slice-value worlds (continued)

and the band phases $z \mapsto e^{I\theta}z$ within a compactified slice; the Lean object S2 is the quotient path category by the direction-composition relations. In particular 0 and N are shared values, not new objects indexed by a zero.

These two geometric views hold the octonionic and slice-value geometry drawn on below.

Lean declaration. [H1](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms H1
'H1' depends on axioms: [proptext, Classical.choice, Quot.sound]
```

Lean declaration. [SphereWorld](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms SphereWorld
'SphereWorld' depends on axioms: [proptext, Classical.choice, Quot.sound]
```

Receipt import. [RECEIPT_PRODUCER_REJECTED: certificate failures: ASection.northState_is_residueActionState_audit, ASection.residueActionState_north_input_audit, ASection.residueActionTransport_north_input_audit, ASection.northRelativeLoop_maps_audit, ASection.northComparison_of_parallelFaces_audit, ASection.relativeActionSquare_transport_audit, ASection.residueTotal_morphism_of_northComparison_audit](#)

40. Definition - The projective base, genuine section functor, and total category

Author's statement

The base is the projective action groupoid

$$\mathcal{B} = PGL(2, \mathbb{R}) \ltimes \text{OnePoint}(\mathbb{R}).$$

Its objects are points of the compactified real projective line and its arrows are projective transformations together with their transport equations.

The matrix construction makes explicit why the analytic object is simultaneously a function and a group element. Let

$$C = \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix}, \quad M(z) = \begin{pmatrix} e^z & 0 \\ 0 & 1 \end{pmatrix}.$$

40. Definition - The projective base, genuine section functor, and total category (continued)

The matrix C is the Cayley map $w \mapsto (w - i)/(w + i)$, and M is exponentiation in matrix form: it takes a complex number z to the diagonal matrix whose entry is e^z . Conjugating an invertible matrix by C and passing to the projective class is a single further operation, used at every level of what follows; for $h \in \text{GL}(2, \mathbb{C})$ write

$$\text{cayleyProjective}(h) = [C h C^{-1}].$$

The passage from a complex number to a slice Möbius map is therefore two steps, exponentiate then conjugate:

$$z \mapsto M(z) \mapsto D(z) = \text{cayleyProjective}(M(z)) \in \text{Möb}(\mathbb{C}^*).$$

Conjugation preserves products, so cayleyProjective is a homomorphism; every later appearance of it — positioning the distinguished element over the base, and acting by a residual stabilizer factor — is this same map applied to a different matrix. Matrix multiplication and $e^{z+w} = e^z e^w$ give

$$M(z + w) = M(z)M(w), \quad D(z + w) = D(z)D(w).$$

Thus exponentiation is the homomorphism which turns addition of logarithmic lifts into composition of the actual slice Möbius maps. In the Lean source these are the chain `diagonalGL`, `diagonalGLHom`, `expUnit`, and `diskExpAction`, with their multiplication laws. Conditions C1–C3 select the A -specific logarithm λ_A of the continued nonzero pole factor and hence the one element

$$D_A = D(\lambda_A) = \text{cayleyProjective}(M(\lambda_A)), \quad M(\lambda_A) = \text{diag}(e^{\lambda_A}, 1) = \text{diag}(u_A, 1).$$

Here λ_A is the logarithm and M is what undoes it: the entry of $M(\lambda_A)$ is $e^{\lambda_A} = u_A$, the multiplier itself, so D_A is an exponential action and not a logarithmic one. The logarithm is the coordinate in which the tape moves; the element it generates lives one exponential above it.

Which logarithm λ_A is, and that it is one at all, is supplied by the slice facts of Part 2 rather than chosen here. The slice exponential is slice-preserving and nowhere zero, with $|\exp q| = e^{\text{Re } q}$ (Theorem 6.1); its logarithm manifold unwraps the multivalued logarithm into a single-valued one, with the branches carried by the manifold and not by the value (Theorem 6.2); and a tame lift along a path is unique once its initial rung is fixed, the ambiguity between branches being exactly a winding rung in $2\pi i\mathbb{Z}$ (Theorem 6.5, Proposition 6.6).

That ambiguity dies under M . Since $e^{\lambda_A + 2\pi i k} = e^{\lambda_A}$, every branch gives the same $M(\lambda_A)$ and hence the same D_A : the logarithm carries the winding, and the exponential action it generates does not see it. This is what lets the tape run through the whole base while the distinguished element stays one element, and it is recorded again below as (B6)–(B8). The Euler and Weierstrass expressions are the two presentations of this same multiplier. The function/group-element passage is the commuting dictionary

$$\begin{array}{ccc} z & \xrightarrow{\text{exp}} & e^z \in \mathbb{C}^\times \\ \downarrow & & \downarrow \\ M(z) = \text{diag}(e^z, 1) & \xrightarrow{\text{cayleyProjective}} & D(z) \in \text{Möb}(\mathbb{C}^*). \end{array}$$

40. Definition - The projective base, genuine section functor, and total category (continued)

The common Euler–Weierstrass multiplier can be followed directly through this dictionary. Write p_A for the pole of A and $g_A = \text{distinguishedPoleFactor } A$. On the punctured pole chart, conditions C2 and C3 give the two identities

$$g_A(z) = (z - p_A) \exp \left(\sum_{p \in A.\iota} \ell_p(z) \right),$$

$$g_A(z) = z^{m_A} R_{\text{fac}, A}(z) e^{h_A(z)} \prod_{n \geq 0} \mathcal{E}(z; q_{A,n}).$$

Condition C1 continues this same factor through the simple pole and proves $u_A = g_A(p_A) \neq 0$. Hence $u_A \in \mathbb{C}^\times$, its logarithm $\lambda_A = \log u_A$ satisfies $e^{\lambda_A} = u_A$, and the distinguished element is

$$D_A = D(\lambda_A) = \text{cayleyProjective}(\text{diag}(u_A, 1)).$$

This calculation also gives its two fixed boundary points. In the standard projective chart $\text{diag}(u_A, 1)$ acts as $w \mapsto u_A w$, which fixes 0 and ∞ and moves everything else. Conjugation carries a fixed point to its Cayley coordinate, so the fixed points of D_A are those of $\text{diag}(u_A, 1)$ read through C : the projective zero and the shared north point N . Therefore

$$D_A(0) = 0, \quad D_A(N) = N,$$

which are the declarations

```
ASection.distinguishedDiskAction.fixes_cayley_zero,
ASection.distinguishedDiskAction.fixes_cayley_N.
```

The Euler expression at 0 and the Weierstrass expression at N are consequently the two boundary readings of this one matrix-built function.

The GPV lift records how that function varies between its boundary presentations. Let $X, Y \in \mathcal{B}$. A projective-base transport from X to Y carries three continuous tapes: a compactified domain path δ^* , a nonzero value path v , its compactification v^* , and a logarithmic lift Γ . Their defining equations are

$$\begin{aligned} \delta^*(0) &= \text{cayleyCoord}(X), & \delta^*(1) &= \text{cayleyCoord}(Y), \\ v^*(t) &= A^*(\delta^*(t)), & e^{\Gamma(t)} &= v(t), \\ \Gamma(1) - \Gamma(0) &= 2\pi i k, & k &\in \mathbb{Z}. \end{aligned}$$

These are precisely the fields of `ASection.GpvTransport`. At each t the lift equation passes through the matrix dictionary:

$$\begin{array}{ccc} \Gamma(t) & \xrightarrow{\text{exp}} & v(t) \in \mathbb{C}^\times \\ \downarrow D & & \downarrow w \mapsto \text{cayleyProjective}(\text{diag}(u, 1)) \\ D(\Gamma(t)) & \xlongequal{\quad} & \text{cayleyProjective}(\text{diag}(v(t), 1)). \end{array}$$

40. Definition - The projective base, genuine section functor, and total category (continued)

Thus every point of the lift is already an action of the same kind as D_A . Taking real parts and norms in the lift equation gives

$$\operatorname{Re} \Gamma(t) = \log \|v(t)\|.$$

The right-hand side shows that the real level is continuous and depends only on the value tape. Any second continuous lift with the same initial rung equals Γ , while every lift of the same value tape has this same real part. Taking real parts in the winding equation gives

$$\operatorname{Re} \Gamma(0) = \operatorname{Re} \Gamma(1),$$

and exponentiating the full equation gives

$$D(\Gamma(0)) = D(\Gamma(1)).$$

The winding integer therefore records the vertical displacement of the logarithmic rung, and the real level and the endpoint Möbius action remain fixed.

Condition C2 now supplies the canonical arithmetic instance of this construction. Along an Euler half-space loop δ ,

$$\Gamma_A(t) = \sum_{p \in A.t} \ell_p(\delta(t)), \quad e^{\Gamma_A(t)} = A(\delta(t)),$$

with the prime index retained inside the complete summation. Local normal convergence makes Γ_A continuous, the zero-free half-space makes its value tape lie in $\mathbb{C}^\times$, and the GPV uniqueness theorem fixes the whole lift once $\Gamma_A(0)$ is chosen. Since $\delta(0) = \delta(1)$, the same prime sum occurs at both endpoints, so its checked winding is $k = 0$. At every instant the square `canonicalAsectionPresentation_euler.toNorth` carries this complete prime tape to the common north frame. Condition C3 supplies the Weierstrass boundary presentation there, and condition C1 identifies both presentations with the continued factor g_A . The prime sum, its value under A , its logarithmic level and winding, and the matrix action it generates are therefore simultaneous coordinates of the distinguished element.

The projective action is read in the direction-indexed sphere-world groupoid. Its objects are directions $I \in S^6$ equipped with their compactified slice spheres, and its morphisms carry the direction and Möbius legs. A normalized value state pairs such a direction with a point of its slice.

We now have to put D_A at every object of the base, and here the choice of an action groupoid pays for itself: no chart is required. In an action groupoid an arrow *is* a group element, and two objects are joined exactly when one carries the other. So the base is a single orbit, every object is reached from the north object N by some element, and whatever D_A becomes at another object is its image under that element — an element of the orbit of D_A , not a new object needing coordinates to describe. The only freedom is which element does the carrying, and fixing that once removes it.

Fix, then, the canonical orbit representative $o_b : N \rightarrow b$, one element carrying N to b , and define the frame at b to be D_A carried along it:

$$a_A(b) = \operatorname{cayleyProjective}(o_b) D_A.$$

40. Definition - The projective base, genuine section functor, and total category (continued)

At the north object itself the representative is the identity, $o_N = 1$, so

$$a_A(N) = \text{cayleyProjective}(1) D_A = D_A.$$

The frame at north *is* the distinguished element, uncarried — which is why every reading taken at N later in this paper is a reading of D_A itself, and why the residue condition can be stated there without a chart.

With the representatives fixed, an arbitrary arrow $f : x \rightarrow y$ factors through N in exactly one way: go back along o_x , do something that fixes N , and come out along o_y . That middle part is the residual north-stabilizer factor

$$r_f = o_y^{-1} f o_x, \quad f = o_y r_f o_x^{-1},$$

and it is the only one, since o_x, o_y are fixed and invertible: $f = o_y r o_x^{-1}$ determines $r = o_y^{-1} f o_x$ outright. Uniqueness of the stabilizer part is therefore the invertibility of the representatives, and every later appeal to it — including at the two boundary transports of the comparison in Lemma 10.6 — is this one line (`GreatCircle.stabilizerPart_unique`).

Call this assignment a_A : it gives each object b the frame $a_A(b)$ above, and each arrow $f : x \rightarrow y$ the Möbius element $a_A(f)$ carrying the frame at x to the frame at y , namely its residual factor conjugated between the two frames. Together,

$$a_A(b) = \text{cayleyProjective}(o_b) D_A, \quad a_A(f) = a_A(y) \text{cayleyProjective}(r_f) a_A(x)^{-1}, \quad (\text{OS})$$

where $f \circ g$ denotes composition in diagrammatic order: f first, then g . Since the representatives are fixed, $r_{1_x} = 1$ and $r_{f \circ g} = r_g r_f$; since `cayleyProjective` is a homomorphism, those two identities are carried to a_A , which is what makes the identity and composition laws hold.

Equation (OS) is the orbit–stabilizer well-definedness of every functor in this paper, and it is the only device used to obtain one. Each is built the same way: the one element D_A , carried to every object along the fixed representatives, and to every arrow by conjugating its residual factor between the frames at the arrow’s two ends. We refer to (OS) by name throughout, and every later positioning statement is an instance of it rather than a new hypothesis.

The preceding construction gives nine checkable readings on the projective base:

- (B1) the C2 Euler and C3 Weierstrass formulas present the one C1-continued factor g_A ;
- (B2) $u_A = g_A(p_A)$ and $\lambda_A = \log u_A$ generate the one matrix action D_A ;
- (B3) the diagonal calculation fixes the two boundary points 0 and N ;
- (B4) the Euler lift retains the complete prime-indexed sum $\sum_{p \in A. \iota} \ell_p$;
- (B5) exponentiation identifies the lift pointwise with the A-value and with its Möbius matrix;
- (B6) the initial rung determines the unique tame continuous lift;
- (B7) its real level is $\log \|A\|$, continuous and independent of the lift branch;
- (B8) its winding lies in $2\pi i\mathbb{Z}$, preserves the endpoint action and real level, and equals zero on the C2 Euler loop;

40. Definition - The projective base, genuine section functor, and total category (continued)

(B9) the unique orbit and stabilizer factors carry every instant of this action through the base, with identity and composition inherited from their factorization laws.

Each line is a consequence of the construction above: (B1)–(B3) come from the common pole factor and its matrix, (B4)–(B8) from its GPV lift, and (B9) from the projective orbit–stabilizer sweep.

The direction-and-Möbius projection is

$$A^{\text{slice}} : \mathcal{B} \longrightarrow \text{SphereWorld},$$

written `AsectionSliceA` in the Lean source. It records the slice sphere and its Möbius transport. The full A-section functor below additionally retains the normalized point and the value produced on that sphere.

The slice-preserving realization on $\mathbb{O}^*$ is the G_2 -equivariant endofunctor $A_{\mathbb{O}} : \mathcal{H}_1 \rightarrow \mathcal{H}_1$. The states it acts on form the *normalized state world* `StateWorldA` (`AsectionStateWorldA`): the action groupoid of G_2 on normalized A-section states. An object is a slice direction $I \in S^6$ together with a normalized coordinate u in the slice that direction selects — a point of the sphere A^{slice} records, now carrying that point and not only the sphere. An arrow is a single element $\gamma \in G_2$, moving the direction and carrying the coordinate with it. The state world has two functors to $\mathcal{H}_1$, reading a state’s coordinate before and after $A_{\mathbb{O}}$ acts: the input eye In_A and the output eye Out_A . The round trip is the checked triangle

$$\begin{array}{ccc} \text{StateWorld}_A & \xrightarrow{\text{In}_A} & \mathcal{H}_1 \\ & \searrow \text{Out}_A & \downarrow A_{\mathbb{O}} \\ & & \mathcal{H}_1, \end{array} \quad \text{Out}_A = \text{In}_A \circ A_{\mathbb{O}}.$$

This equality is `AsectionState_input_then_equivariant`: it holds on arrows as well as objects, so the normalized input coordinates genuinely drive the realized A-values.

Positioning this round trip over $b \in \mathcal{B}$ forms a constrained graph. An object in the fibre consists of an input state x , the forced positioned state $a_A(b) \cdot x$, and the forced value $\text{Out}_A(a_A(b) \cdot x)$. Thus both the positioned point and its value are determined by the input and the action. For $f : b \rightarrow b'$, the two legs of the orbit–stabilizer square transport the input and positioned faces, and its already-proved commuting equation makes the target graph well-defined. Reading that same square through In_A and Out_A gives the checked input and output naturality squares. Identity and composition of these graph transports are inherited from the unique factorization of r_f . Sweeping the whole constrained graph through the base therefore gives the genuine A-section functor

$$\mathcal{A}_A : \mathcal{B} \longrightarrow \mathbf{Grpd},$$

written `AsectionActionDiagramA` in the Lean source.

Both its source and its targets are action groupoids. The source $\mathcal{B} = \text{PGL}(2, \mathbb{R}) \ltimes \text{OnePoint}(\mathbb{R})$ is the projective action groupoid: an object is a point of the compactified real projective line, and an arrow $f : x \rightarrow y$ is a projective transformation together with its transport equation, so it is one element of $\text{PGL}(2, \mathbb{R})$ carrying x to y . Each groupoid it produces is an action groupoid of G_2 on states, and the element positioned there is the distinguished one, D_A .

40. Definition - The projective base, genuine section functor, and total category (continued)

On a base object b , the groupoid $\mathcal{A}_A(b)$ consists of the normalized A-section states carrying D_A to that object. The carrying is done by the object frame $a_A(b) = \text{cayleyProjective}(o_b)D_A$, which is D_A conjugated along the canonical orbit representative o_b — the same element at every base object, moved into position. An object of $\mathcal{A}_A(b)$ is one constrained graph (A): a slice direction $I \in S^6$ and a normalized input coordinate u , together with the positioned coordinate $a_A(b) \cdot (I, u)$ that this frame determines and the A-value realized there. An arrow is a single element $\gamma \in G_2$ acting on the slice direction and carrying the rest of the graph with it, so $\mathcal{A}_A(b)$ is the action groupoid of G_2 on those states.

On a base arrow $f : x \rightarrow y$ — that is, on one projective transformation — the value $\mathcal{A}_A(f)$ is transport by the arrow element

$$a_A(f) = a_A(y) \text{cayleyProjective}(r_f) a_A(x)^{-1}$$

of (OS): the distinguished action read at the target frame, composed with the Cayley action of the residual factor r_f that f leaves after its orbit part is stripped, composed with the inverse of the distinguished action read at the source frame. On a state this moves the input coordinate by the input leg $\text{cayleyProjective}(r_f)$ and the positioned coordinate by the positioned leg, the two horizontal legs of the square whose verticals are those two frames. On an arrow it is the identity on γ , since G_2 acts on the slice direction while $a_A(f)$ acts on the coordinate, so the two commute. The identity and composition laws hold because the fixed representatives force $r_{1_x} = 1$ and $r_{f \circ g} = r_g r_f$, and $\text{cayleyProjective}(-)$ is a homomorphism.

The nine base readings (B1)–(B9) pass through this constrained graph by the certified projective/GPV action squares. Three further readings are thereby realized in the generated states. First, condition C3 supplies the populated residue- $\mathbb{C}$ sphere states, represented by `residueActionState` and certified by `residueActionState_mem`. Second, condition C4 makes their population infinite through `c4_infinite`. Third, every such state carries its intrinsic real coordinate through `transportLevel`. These are the three output readings. Together with (B1)–(B9) they form the $9 + 3$ partition: nine properties of the one distinguished action on the base, followed functorially by three readings of the A-generated residue states in the total. The total value-transport category is

$$\mathcal{T}_A = \int_B \mathcal{A}_A.$$

An object is a base position together with a normalized A-section value state in its fibre. A morphism is a base channel together with the compatible A-section transport supplied by A . The populated residue- $\mathbb{C}$ zero-sphere states are the degenerate-fibre outputs of this completed construction, produced by C3–C4. The associated diagram $\pi_0 \circ \mathcal{A}_A : \mathcal{B} \rightarrow \mathbf{Set}$ is the components diagram required by the Grothendieck/colimit readout (Lemma 10.3).

Lean declaration. `GreatCircle.Base`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

40. Definition - The projective base, genuine section functor, and total category (continued)

Lean command and literal axiom print.

```
#print axioms GreatCircle.Base
'GreatCircle.Base' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. [GreatCircle.cayleyGL](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms GreatCircle.cayleyGL
'GreatCircle.cayleyGL' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. [GreatCircle.diagonalGL](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms GreatCircle.diagonalGL
'GreatCircle.diagonalGL' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. [GreatCircle.diagonalGL_mul](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms GreatCircle.diagonalGL_mul
'GreatCircle.diagonalGL_mul' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [GreatCircle.diagonalGLHom](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms GreatCircle.diagonalGLHom
'GreatCircle.diagonalGLHom' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

40. Definition - The projective base, genuine section functor, and total category (continued)

Lean declaration. [GreatCircle.expUnit](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms GreatCircle.expUnit
'GreatCircle.expUnit' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. [GreatCircle.expUnit_add](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms GreatCircle.expUnit_add
'GreatCircle.expUnit_add' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. [GreatCircle.diskExpAction](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms GreatCircle.diskExpAction
'GreatCircle.diskExpAction' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [GreatCircle.diskExpAction_add](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms GreatCircle.diskExpAction_add
'GreatCircle.diskExpAction_add' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [GreatCircle.cayleyProjective](#)

Author's object.

40. Definition - The projective base, genuine section functor, and total category (continued)

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms GreatCircle.cayleyProjective
'GreatCircle.cayleyProjective' depends on axioms: [propept, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [GreatCircle.cayleyProjective_mul](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms GreatCircle.cayleyProjective_mul
'GreatCircle.cayleyProjective_mul' depends on axioms: [propept, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [GreatCircle.orbitRep](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms GreatCircle.orbitRep
'GreatCircle.orbitRep' depends on axioms: [propept, Classical.choice, Quot.sound]
```

Lean declaration. [GreatCircle.orbitRep_spec](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms GreatCircle.orbitRep_spec
'GreatCircle.orbitRep_spec' depends on axioms: [propept, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [GreatCircle.stabilizerPart](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

40. Definition - The projective base, genuine section functor, and total category (continued)

```
#print axioms GreatCircle.stabilizerPart
'GreatCircle.stabilizerPart' depends on axioms: [propept, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [GreatCircle.orbit_stabilizer_factor](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms GreatCircle.orbit_stabilizer_factor
'GreatCircle.orbit_stabilizer_factor' depends on axioms: [propept, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [GreatCircle.stabilizerPart_unique](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms GreatCircle.stabilizerPart_unique
'GreatCircle.stabilizerPart_unique' depends on axioms: [propept, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [GreatCircle.stabilizerPart_id](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms GreatCircle.stabilizerPart_id
'GreatCircle.stabilizerPart_id' depends on axioms: [propept, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [GreatCircle.stabilizerPart_comp](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

40. Definition - The projective base, genuine section functor, and total category (continued)

```
#print axioms GreatCircle.stabilizerPart_comp
'GreatCircle.stabilizerPart_comp' depends on axioms: [propept, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [ASection.distinguishedDiskAction](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.distinguishedDiskAction
'ASection.distinguishedDiskAction' depends on axioms: [propept, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [ASection.distinguishedDiskAction_eq_fullMultiplier](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.distinguishedDiskAction_eq_fullMultiplier
'ASection.distinguishedDiskAction_eq_fullMultiplier' depends on axioms: [propept,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [ASection.distinguishedPoleFactor_euler](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.distinguishedPoleFactor_euler
'ASection.distinguishedPoleFactor_euler' depends on axioms: [propept,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [ASection.distinguishedPoleFactor_weierstrass](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

40. Definition - The projective base, genuine section functor, and total category (continued)

```
#print axioms ASection.distinguishedPoleFactor_weierstrass
'ASection.distinguishedPoleFactor_weierstrass' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [ASection.eulerPrimeSum](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.eulerPrimeSum
'ASection.eulerPrimeSum' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. [ASection.eulerDiskAction_eq_value](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.eulerDiskAction_eq_value
'ASection.eulerDiskAction_eq_value' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [ASection.GpvTransport](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.GpvTransport
'ASection.GpvTransport' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. [ASection.GpvTransport.diskExpAction_eq_value](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.GpvTransport.diskExpAction_eq_value
'ASection.GpvTransport.diskExpAction_eq_value' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```


40. Definition - The projective base, genuine section functor, and total category (continued)

Lean declaration. [ASection.GpvTransport.diskExpAction_endpoint_eq](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.GpvTransport.diskExpAction_endpoint_eq
'ASection.GpvTransport.diskExpAction_endpoint_eq' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [ASection.GpvTransport.lift_endpoint_re_eq](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.GpvTransport.lift_endpoint_re_eq
'ASection.GpvTransport.lift_endpoint_re_eq' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [ASection.GpvTransport.level](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.GpvTransport.level
'ASection.GpvTransport.level' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [ASection.GpvTransport.continuous_level](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.GpvTransport.continuous_level
'ASection.GpvTransport.continuous_level' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

40. Definition - The projective base, genuine section functor, and total category (continued)

Lean declaration. [ASection.GpvTransport.lift_unique](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.GpvTransport.lift_unique
'ASection.GpvTransport.lift_unique' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [ASection.GpvTransport.level_independent](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.GpvTransport.level_independent
'ASection.GpvTransport.level_independent' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [ASection.GpvTransport.value_at_source](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.GpvTransport.value_at_source
'ASection.GpvTransport.value_at_source' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [ASection.GpvTransport.value_at_target](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.GpvTransport.value_at_target
'ASection.GpvTransport.value_at_target' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

40. Definition - The projective base, genuine section functor, and total category (continued)

Lean declaration. [ASection.GpvTransport.endpoint_log_norm_eq](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.GpvTransport.endpoint_log_norm_eq
'ASection.GpvTransport.endpoint_log_norm_eq' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [ASection.GpvTransport.endpoint_norm_eq](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.GpvTransport.endpoint_norm_eq
'ASection.GpvTransport.endpoint_norm_eq' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [ASection.projectiveGpvActionSquare](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.projectiveGpvActionSquare
'ASection.projectiveGpvActionSquare' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [ASection.projectiveGpvActionSquare_level](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.projectiveGpvActionSquare_level
'ASection.projectiveGpvActionSquare_level' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

40. Definition - The projective base, genuine section functor, and total category (continued)

Lean declaration. [ASection.projectiveGpvActionSquare_input_commutes](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.projectiveGpvActionSquare_input_commutes
'ASection.projectiveGpvActionSquare_input_commutes' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [ASection.projectiveGpvActionSquare_output_commutes](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.projectiveGpvActionSquare_output_commutes
'ASection.projectiveGpvActionSquare_output_commutes' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [ASection.canonicalAsectionPresentation_euler_prime_stack](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.canonicalAsectionPresentation_euler_prime_stack
'ASection.canonicalAsectionPresentation_euler_prime_stack' depends on axioms:
↪ [propext, Classical.choice, Quot.sound]
```

Lean declaration. [ASection.canonicalAsectionPresentation_euler_lift_exp](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.canonicalAsectionPresentation_euler_lift_exp
'ASection.canonicalAsectionPresentation_euler_lift_exp' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

40. Definition - The projective base, genuine section functor, and total category (continued)

Lean declaration. [ASection.canonicalAsectionPresentation_euler_lift_unique](#)
Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.canonicalAsectionPresentation_euler_lift_unique
'ASection.canonicalAsectionPresentation_euler_lift_unique' depends on axioms:
↪ [propext, Classical.choice, Quot.sound]
```

Lean declaration. [ASection.canonicalAsectionPresentation_euler_level_continuous](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.canonicalAsectionPresentation_euler_level_continuous
'ASection.canonicalAsectionPresentation_euler_level_continuous' depends on axioms:
↪ [propext,
   Classical.choice,
   Quot.sound]
```

Lean declaration. [ASection.canonicalAsectionPresentation_euler_winding](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.canonicalAsectionPresentation_euler_winding
'ASection.canonicalAsectionPresentation_euler_winding' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [ASection.canonicalAsectionPresentation_euler_toNorth](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.canonicalAsectionPresentation_euler_toNorth
'ASection.canonicalAsectionPresentation_euler_toNorth' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

40. Definition - The projective base, genuine section functor, and total category (continued)

Lean declaration. [ASection.projectiveObjectFrame](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.projectiveObjectFrame
'ASection.projectiveObjectFrame' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [ASection.projectiveArrowElement](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.projectiveArrowElement
'ASection.projectiveArrowElement' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [ASection.projectiveObjectFrame_north](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.projectiveObjectFrame_north
'ASection.projectiveObjectFrame_north' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [ASection.distinguishedDiskAction_fixes_cayley_zero](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.distinguishedDiskAction_fixes_cayley_zero
'ASection.distinguishedDiskAction_fixes_cayley_zero' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

40. Definition - The projective base, genuine section functor, and total category (continued)

Lean declaration. [ASection.distinguishedDiskAction_fixes_cayley_N](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.distinguishedDiskAction_fixes_cayley_N
'ASection.distinguishedDiskAction_fixes_cayley_N' depends on axioms: [propept,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [ASection.projectiveArrowElement_id](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.projectiveArrowElement_id
'ASection.projectiveArrowElement_id' depends on axioms: [propept, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [ASection.projectiveArrowElement_comp](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.projectiveArrowElement_comp
'ASection.projectiveArrowElement_comp' depends on axioms: [propept,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [ASection.AsectionSlice](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.AsectionSlice
'ASection.AsectionSlice' depends on axioms: [propept, Classical.choice, Quot.sound]
```

Lean declaration. [ASection.AsectionEquivariant](#)

Author's object.

40. Definition - The projective base, genuine section functor, and total category (continued)

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.AsectionEquivariant
'ASection.AsectionEquivariant' depends on axioms: [proptext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [ASection.AsectionStateInput](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.AsectionStateInput
'ASection.AsectionStateInput' depends on axioms: [proptext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [ASection.AsectionStateOutput](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.AsectionStateOutput
'ASection.AsectionStateOutput' depends on axioms: [proptext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [ASection.AsectionState_input_then_equivariant](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.AsectionState_input_then_equivariant
'ASection.AsectionState_input_then_equivariant' depends on axioms: [proptext,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [ASection.ActionTransportSquare.coordinateTransport_commutates](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

40. Definition - The projective base, genuine section functor, and total category (continued)

```
#print axioms ASection.ActionTransportSquare.coordinateTransport_commutes
'ASection.ActionTransportSquare.coordinateTransport_commutes' depends on axioms:
↪ [propeft, Classical.choice, Quot.sound]
```

Lean declaration. [ASection.ActionTransportSquare.input_commutes](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.ActionTransportSquare.input_commutes
'ASection.ActionTransportSquare.input_commutes' depends on axioms: [propeft,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [ASection.ActionTransportSquare.output_commutes](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.ActionTransportSquare.output_commutes
'ASection.ActionTransportSquare.output_commutes' depends on axioms: [propeft,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [ASection.AsectionActionState](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.AsectionActionState
'ASection.AsectionActionState' depends on axioms: [propeft, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [ASection.AsectionActionTransport](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

40. Definition - The projective base, genuine section functor, and total category (continued)

```
#print axioms ASection.AsectionActionTransport
'ASection.AsectionActionTransport' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [ASection.AsectionActionState.ofInput](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.AsectionActionState.ofInput
'ASection.AsectionActionState.ofInput' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [ASection.AsectionActionState_ofInput_value](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.AsectionActionState_ofInput_value
'ASection.AsectionActionState_ofInput_value' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [ASection.AsectionActionTransport_obj_input](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.AsectionActionTransport_obj_input
'ASection.AsectionActionTransport_obj_input' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [ASection.AsectionActionTransport_obj_positioned](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

40. Definition - The projective base, genuine section functor, and total category (continued)

```
#print axioms ASection.AsectionActionTransport_obj_positioned
'ASection.AsectionActionTransport_obj_positioned' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [ASection.AsectionActionTransport_obj_value](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.AsectionActionTransport_obj_value
'ASection.AsectionActionTransport_obj_value' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [ASection.AsectionActionTransport_id](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.AsectionActionTransport_id
'ASection.AsectionActionTransport_id' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [ASection.AsectionActionTransport_comp](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.AsectionActionTransport_comp
'ASection.AsectionActionTransport_comp' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [ASection.AsectionActionDiagram](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

40. Definition - The projective base, genuine section functor, and total category (continued)

```
#print axioms ASection.AsectionActionDiagram
'ASection.AsectionActionDiagram' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [ASection.residueActionState](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.residueActionState
'ASection.residueActionState' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [ASection.residueActionState_mem](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.residueActionState_mem
'ASection.residueActionState_mem' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [ASection.c4_infinite](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.c4_infinite
'ASection.c4_infinite' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. [ASection.transportLevel](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.transportLevel
'ASection.transportLevel' depends on axioms: [propext, Classical.choice, Quot.sound]
```

40. Definition - The projective base, genuine section functor, and total category (continued)

Lean declaration. [ASection.ambientTotalCategory](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.ambientTotalCategory
'ASection.ambientTotalCategory' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [ASection.ambientTotalGroupoid](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.ambientTotalGroupoid
'ASection.ambientTotalGroupoid' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [grothendieckGrpdGroupoid](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms grothendieckGrpdGroupoid
'grothendieckGrpdGroupoid' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [CategoryTheory.Grothendieck](#)

Author's object.

Object relation. Object relation not checked

Exact state. External library dependency; approved kernel-reported axiom surface; no authored-object anchor required

Lean command and literal axiom print.

```
#print axioms CategoryTheory.Grothendieck
'CategoryTheory.Grothendieck' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

40. Definition - The projective base, genuine section functor, and total category (continued)

Receipt import. `RECEIPT_PRODUCER_REJECTED`: certificate failures: `ASection.northState_is_residueActionState_audit`, `ASection.residueActionState_north_input_audit`, `ASection.residueActionTransport_north_input_audit`, `ASection.northRelativeLoop_maps_audit`, `ASection.northComparison_of_parallelFaces_audit`, `ASection.relativeActionSquare_transport_audit`, `ASection.residueTotal_morphism_of_northComparison_audit`

41. Proposition - Slice-regular Weierstrass factorization; the content of C3

Author's statement

Let $A \in \mathcal{R}$ have a zero of order $m \geq 0$ at the origin and no non-real isolated zeros (for $A \in \mathcal{R}$ the latter is automatic: non-real zeros come in whole spheres, Lemma 10.2). Then on $\mathbb{O}^* \setminus \{N\}$ it factors as

$$A = q^m R S h,$$

where q denotes the octonionic coordinate function (the identity section, so q^m carries the order- m zero at the origin), R is a slice-preserving factor vanishing exactly at the real zeros of A , S is a factor vanishing exactly at its spherical (residue- $\mathbb{C}$) zero-spheres, and h is never vanishing; each of R, S is the slice-regular Weierstrass product over the corresponding zeros. The factorization holds *including the case where either family of zeros is infinite*.

Lean declaration. `weierstrasseE`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms weierstrasseE
'weierstrasseE' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. `spherePrimary`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms spherePrimary
'spherePrimary' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. `ASection.c3_multipliable`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

41. Proposition - Slice-regular Weierstrass factorization; the content of C3 (continued)

Lean command and literal axiom print.

```
#print axioms ASection.c3_multipliable
'ASection.c3_multipliable' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [ASection.c3_locMajorant](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.c3_locMajorant
'ASection.c3_locMajorant' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. [ASection.c3_factorization](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.c3_factorization
'ASection.c3_factorization' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Receipt import. RECEIPT_PRODUCER_REJECTED: certificate failures: ASection.northState_is_residueActionState_audit, ASection.residueActionState_north_input_audit, ASection.residueActionTransport_north_input_audit, ASection.northRelativeLoop_maps_audit, ASection.northComparison_of_parallelFaces_audit, ASection.relativeActionSquare_transport_audit, ASection.residueTotal_morphism_of_northComparison_audit

42. Lemma - Residue- $\mathbb{C}\mathbb{C}$ zeros are 6-sphere components of $\mathcal{H}_1$

Author's statement

For $A \in \mathcal{R}$, the non-real zeros of A are 6-spheres, each a single connected component of the translation groupoid $\mathcal{H}_1$.

Lean declaration. [ASection.sphereZero_mem_CResidueZeroLocus](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

42. Lemma - Residue- $\mathbb{C}\mathbb{C}$ zeros are $\mathbb{C}$ -sphere components of $\mathcal{H}_1$ (continued)

Lean command and literal axiom print.

```
#print axioms ASection.sphereZero_mem_CResidueZeroLocus
'ASection.sphereZero_mem_CResidueZeroLocus' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [ASection.mem_CResidueZeroLocus_iff_exists_sphereZero](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.mem_CResidueZeroLocus_iff_exists_sphereZero
'ASection.mem_CResidueZeroLocus_iff_exists_sphereZero' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [G2.exists_smul_eq_of_mem_unitImaginarySphere](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms G2.exists_smul_eq_of_mem_unitImaginarySphere
'G2.exists_smul_eq_of_mem_unitImaginarySphere' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

Receipt import. RECEIPT_PRODUCER_REJECTED: certificate failures: ASection.northState_is_residueActionState_audit, ASection.residueActionState_north_input_audit, ASection.residueActionTransport_north_input_audit, ASection.northRelativeLoop_maps_audit, ASection.northComparison_of_parallelFaces_audit, ASection.relativeActionSquare_transport_audit, ASection.residueTotal_morphism_of_northComparison_audit

43. Lemma - π_0 of a Grothendieck construction

Author's statement

For a functor $F : \mathcal{B} \rightarrow \mathbf{Grpd}$, the connected-components functor carries the Grothendieck construction to the colimit of the component diagram:

$$\pi_0\left(\int_{\mathcal{B}} F\right) \cong \operatorname{colim}_{\mathcal{B}} (\pi_0 \circ F).$$

43. Lemma - π_0 of a Grothendieck construction (continued)

Lean declaration. `CategoryTheory.Grothendieck`

Author's object.

Object relation. Object relation not checked

Exact state. External library dependency; approved kernel-reported axiom surface; no authored-object anchor required

Lean command and literal axiom print.

```
#print axioms CategoryTheory.Grothendieck
'CategoryTheory.Grothendieck' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. `pi0GrothendieckEquiv`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms pi0GrothendieckEquiv
'pi0GrothendieckEquiv' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Lean declaration. `pi0_grothendieck`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms pi0_grothendieck
'pi0_grothendieck' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Receipt import. `RECEIPT_PRODUCER.REJECTED`: certificate failures: `ASection.northState_is_residueActionState_audit`, `ASection.residueActionState_north_input_audit`, `ASection.residueActionTransport_north_input_audit`, `ASection.northRelativeLoop_maps_audit`, `ASection.northComparison_of_parallelFaces_audit`, `ASection.relativeActionSquare_transport_audit`, `ASection.residueTotal_morphism_of_northComparison_audit`

44. Definition - $\mathcal{A}$ -sections

Author's statement

An A -section is a section A of the ring $\mathcal{R}$ of slice-preserving slice-regular functions on $\mathbb{O}^* = S^8$ (Definition 5.1) with the following four properties.

- (C1) *Meromorphic continuation; single simple pole.* A is meromorphically continued to $\mathbb{O}^*$ with exactly one pole; it is simple, lies at a real point, and its value there is the point

44. Definition - $\mathbb{A}$ -sections (continued)

at infinity $\infty = N$.

- (C2) *Infinite Euler product.* On a slice right half-space $\Omega_0 \subset \mathbb{O}^*$, $A = \exp(\sum_p \ell_p)$ for an *infinite* family $\{\ell_p\}$ of slice-preserving slice-regular functions each zero-free on Ω_0 , the family summable *locally normally* on Ω_0 (the sense in which the cited Euler products converge; Theorem 1.2); in particular A is zero-free on Ω_0 .
- (C3) *Infinite Weierstrass factorization.* Over the *full divisor* of A , including the single pole of (C1), at the real point p_0 , multiplicity -1 , carried by the explicit left factor, on $\mathbb{O}^* \setminus \{p_0\}$,

$$(q - p_0) A = q^m R e^g \prod_n \mathcal{E}(\cdot; q_n),$$

where q is the octonionic coordinate, $m \geq 0$, R is the slice-regular Weierstrass product over the *nonzero* residue- $\mathbb{R}$ (real) zeros of A , vanishing only on $\mathbb{R}$, g is slice-preserving entire, $\{q_n\}$ enumerates the residue- $\mathbb{C}$ zero-spheres of A , and $\mathcal{E}(\cdot; q_n)$ is the slice-regular Weierstrass primary factor of q_n (Proposition 10.1; [2, Prop. 3.1, Thm. 3.2], [10]); the factorization holds with either family infinite, the product converging *locally normally* on $\mathbb{O}^* \setminus \{p_0\}$ (the convergence of Proposition 10.1). (The left factor makes “full divisor” literal: the divisor of a meromorphic section includes its single pole; classically, Hadamard factors $(s - 1)\zeta(s)$.)

- (C4) *Infinitely many residue- $\mathbb{C}$ zeros.* The index set $\{q_n\}$ is infinite. A closed zero of such a section is a real point (residue field $\mathbb{R}$) or a 6-sphere $S_{(\sigma, \gamma)}$ (residue field $\mathbb{C}$; [2]); residue- $\mathbb{C}$ names the classically nontrivial zeros (Part 2).

Lean declaration. [ASection](#)

Author’s object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection
'ASection' depends on axioms: [proptext, Classical.choice, Quot.sound]
```

Lean declaration. [weierstrasse](#)

Author’s object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms weierstrasse
'weierstrasse' depends on axioms: [proptext, Classical.choice, Quot.sound]
```

Lean declaration. [spherePrimary](#)

44. Definition - $\mathcal{A}$ -sections (continued)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms spherePrimary
'spherePrimary' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Receipt import. `RECEIPT_PRODUCER_REJECTED`: certificate failures: `ASection.northState_is_residueActionState_audit`, `ASection.residueActionState_north_input_audit`, `ASection.residueActionTransport_north_input_audit`, `ASection.northRelativeLoop_maps_audit`, `ASection.northComparison_of_parallelFaces_audit`, `ASection.relativeActionSquare_transport_audit`, `ASection.residueTotal_morphism_of_northComparison_audit`

45. Definition - The residue subdiagram and its inclusion

Author's statement

The *residue locus* of an $\mathcal{A}$ -section is $\{z : A(z) = 0, \text{Im } z > 0\}$ (`CResidueZeroLocus`). The *residue subdiagram* $\mathcal{R}_A : \mathcal{B} \rightarrow \mathbf{Grpd}$ (`AsectionCResidueDiagram`) is the preimage of the states this locus selects at the north frame — a preimage taken in the total action groupoid, not of a set — with the witnessing base arrow carried as data.

It has the same source as $\mathcal{A}_A$, the projective action groupoid $\mathcal{B}$, and its values are again action groupoids of G_2 — the selection cuts down which states are present without altering what acts on them.

On a base object X , the value $\mathcal{R}_A(X)$ is the full subgroupoid of $\mathcal{A}_A(X)$ on the witnessed states. An object there is a state x of $\mathcal{A}_A(X)$ together with a north residue state x_N and a base arrow $g : N \rightarrow X$ — one projective transformation carrying the north point to X — satisfying $\mathcal{A}_A(g)(x_N) = x$; the witness travels with the object. An arrow is an arrow of $\mathcal{A}_A(X)$ between two such states, so again a single element $\gamma \in G_2$: the selection is on objects, and G_2 still acts.

On a base arrow $f : X \rightarrow Y$, the value $\mathcal{R}_A(f)$ is the restriction of the ambient transport $\mathcal{A}_A(f)$, which is transport by the arrow element $a_A(f)$ carrying D_A between the frames at X and Y . That restriction lands in the subgroupoid because a witnessed state stays witnessed, with the *same* north state and the composite witness $g \circ f$: functoriality of $\mathcal{A}_A$ gives $\mathcal{A}_A(g \circ f)(x_N) = \mathcal{A}_A(f)(\mathcal{A}_A(g)(x_N)) = \mathcal{A}_A(f)(x)$. So the north state is what is preserved, and the witnessing projective transformation is what accumulates. Its inclusion $\iota_A : \mathcal{R}_A \Rightarrow \mathcal{A}_A$ (`AsectionCResidueInclusion`) sends a witnessed object to its ambient state and is the identity on arrows; it is fibrewise fully faithful onto that inverse image, and its naturality squares retain the ambient $\mathcal{A}$ -section transport itself.

Lean declaration. `ASection.CResidueZeroLocus`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor

45. Definition - The residue subdiagram and its inclusion (continued)

unratified

Lean command and literal axiom print.

```
#print axioms ASection.CResidueZeroLocus
'ASection.CResidueZeroLocus' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [ASection.AsectionCResidueDiagram](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.AsectionCResidueDiagram
'ASection.AsectionCResidueDiagram' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [ASection.AsectionCResidueInclusion](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.AsectionCResidueInclusion
'ASection.AsectionCResidueInclusion' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [ASection.AsectionCResidueInclusionTotal](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.AsectionCResidueInclusionTotal
'ASection.AsectionCResidueInclusionTotal' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [ASection.AsectionCResidueInclusionTotal_full](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

45. Definition - The residue subdiagram and its inclusion (continued)

```
#print axioms ASection.AsectionCResidueInclusionTotal_full
'ASection.AsectionCResidueInclusionTotal_full' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [ASection.AsectionCResidueInclusionTotal_faithful](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.AsectionCResidueInclusionTotal_faithful
'ASection.AsectionCResidueInclusionTotal_faithful' depends on axioms: [propext,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [ASection.residueTotalCategory](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.residueTotalCategory
'ASection.residueTotalCategory' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [ASection.residueTotalGroupoid](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.residueTotalGroupoid
'ASection.residueTotalGroupoid' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
```

Receipt import. [RECEIPT_PRODUCER_REJECTED](#): certificate failures: [ASection.northState_is_residueActionState_audit](#), [ASection.residueActionState_north_input_audit](#), [ASection.residueActionTransport_north_input_audit](#), [ASection.northRelativeLoop_maps_audit](#), [ASection.northComparison_of_parallelFaces_audit](#), [ASection.relativeActionSquare_transport_audit](#), [ASection.residueTotal_morphism_of_northComparison_audit](#)

46. Lemma - Transitivity of the C-residue total

Author's statement

For an A-section A , the total inverse-image groupoid

$$\int_B \mathcal{R}_A$$

is transitive: any two of its objects are joined by a morphism.

Lean declaration. [ASection.sweepTransitive_on_residueSystem](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted with sorryAx; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.sweepTransitive_on_residueSystem
'ASection.sweepTransitive_on_residueSystem' depends on axioms: [propeXt, sorryAx,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [ASection.northRelativeLoop_maps](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.northRelativeLoop_maps
'ASection.northRelativeLoop_maps' depends on axioms: [propeXt, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [ASection.northComparison_of_parallelFaces](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.northComparison_of_parallelFaces
'ASection.northComparison_of_parallelFaces' depends on axioms: [propeXt,
↪ Classical.choice, Quot.sound]
```

Receipt import. RECEIPT_PRODUCER_REJECTED: certificate failures: ASection.northState_is_residueActionState_audit, ASection.residueActionState.north_input_audit, ASection.residueActionTransport.north_input_audit, ASection.northRelativeLoop_maps_audit, ASection.northComparison_of_parallelFaces_audit, ASection.relativeActionSquare_transport_audit, ASection.residueTotal_morphism_of_northComparison_audit

47. Theorem - Concentricity

Author's statement

Let A be an A -section. The infinitely many C -residue zero S^6 spheres transported by the A -section functor $\mathcal{A}_A : \mathcal{B} \rightarrow \mathbf{Grpd}$, from the projective great-circle base $\mathcal{B}$ to the A -generated normalized value groupoids, form exactly one colimit/component class, and that singleton class carries one real value

$$c \in \mathbb{R}.$$

Consequently every populated residue- $\mathbb{C}$ zero 6-sphere has centre c , so the infinitely many populated residue- $\mathbb{C}$ zero spheres are concentric.

Lean declaration. [ASection.concentricity](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted with sorryAx; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.concentricity
'ASection.concentricity' depends on axioms: [propext, sorryAx, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [ASection.residueTotal_isConnected](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted with sorryAx; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.residueTotal_isConnected
'ASection.residueTotal_isConnected' depends on axioms: [propext, sorryAx,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [ASection.residueTotal_pi0_singleton](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted with sorryAx; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms ASection.residueTotal_pi0_singleton
'ASection.residueTotal_pi0_singleton' depends on axioms: [propext, sorryAx,
↪ Classical.choice, Quot.sound]
```

Lean declaration. [ASection.transportLevel](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unratified

Lean command and literal axiom print.

47. Theorem - Concentricity (continued)

```
#print axioms ASection.transportLevel
'ASection.transportLevel' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Receipt import. **RECEIPT_PRODUCER_REJECTED:** certificate failures: ASection.northState_is_residueActionState_audit, ASection.residueActionState_north_input_audit, ASection.residueActionTransport_north_input_audit, ASection.northRelativeLoop_maps_audit, ASection.northComparison_of_parallelFaces_audit, ASection.relativeActionSquare_transport_audit, ASection.residueTotal_morphism_of_northComparison_audit

48. Corollary - Translation to the classical framework

Author's statement

Under the residue- $\mathbb{C}$ /classically-nontrivial-zero identification, the already-proved populated residue- $\mathbb{C}$ zero spheres are the classically nontrivial zero spheres and retain the common centre c supplied by Theorem 10.7.

Lean declaration. `ASection.nontrivial_one_centre`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted with sorryAx; authored-object anchor unrati

Lean command and literal axiom print.

```
#print axioms ASection.nontrivial_one_centre
'ASection.nontrivial_one_centre' depends on axioms: [propext, sorryAx,
↪ Classical.choice, Quot.sound]
```

Receipt import. **RECEIPT_PRODUCER_REJECTED:** certificate failures: ASection.northState_is_residueActionState_audit, ASection.residueActionState_north_input_audit, ASection.residueActionTransport_north_input_audit, ASection.northRelativeLoop_maps_audit, ASection.northComparison_of_parallelFaces_audit, ASection.relativeActionSquare_transport_audit, ASection.residueTotal_morphism_of_northComparison_audit

49. Corollary - $\zeta_{\mathbb{O}^*}$ is an A -section

Author's statement

$\zeta_{\mathbb{O}^*}$ is an A -section (Definition 10.4): a section of $\mathcal{R}$ satisfying (C1)–(C4).

Lean declaration. `zetaSection`

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted; approved axiom surface; authored-object anchor unrati

Lean command and literal axiom print.

```
#print axioms zetaSection
```


49. Corollary - ζ is an $\mathcal{A}$ -section (continued)

'zetaSection' depends on axioms: [propext, Classical.choice, Quot.sound]

Receipt import. **RECEIPT_PRODUCER_REJECTED:** certificate failures: ASection.northState.is_residueActionState_audit, ASection.residueActionState_north_input_audit, ASection.residueActionTransport_north_input_audit, ASection.northRelativeLoop_maps_audit, ASection.northComparison_of_parallelFaces_audit, ASection.relativeActionSquare_transport_audit, ASection.residueTotal_morphism_of_northComparison_audit

50. Corollary - Riemann Hypothesis

Author's statement

Every nontrivial zero of the classical Riemann zeta function has real part $\frac{1}{2}$; in particular infinitely many zeros lie on the line $\Re s = \frac{1}{2}$.

Lean declaration. [zeta_riemannHypothesis](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted with sorryAx; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms zeta_riemannHypothesis
'zeta_riemannHypothesis' depends on axioms: [propext, sorryAx, Classical.choice,
↪ Quot.sound]
```

Lean declaration. [zeta_criticalLine_zeros_infinite](#)

Author's object.

Object relation. Object relation not checked

Exact state. Lean declaration accepted with sorryAx; authored-object anchor unratified

Lean command and literal axiom print.

```
#print axioms zeta_criticalLine_zeros_infinite
'zeta_criticalLine_zeros_infinite' depends on axioms: [propext, sorryAx,
↪ Classical.choice, Quot.sound]
```

Receipt import. **RECEIPT_PRODUCER_REJECTED:** certificate failures: ASection.northState.is_residueActionState_audit, ASection.residueActionState_north_input_audit, ASection.residueActionTransport_north_input_audit, ASection.northRelativeLoop_maps_audit, ASection.northComparison_of_parallelFaces_audit, ASection.relativeActionSquare_transport_audit, ASection.residueTotal_morphism_of_northComparison_audit